

On the Regularity of Singular Set

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1 Introduction

This reading report is on the three papers [2, 3, 4] of A. Figalli et al. They made use of various monotonicity formulae to investigate higher order blow-ups of the solution of the obstacle problem, and proved some results on the regularity of the singular set.

Throughout this article, u always denotes a solution to the obstacle problem. That is, $u \geq 0$ and satisfies

$$\Delta u = \chi_{\{u > 0\}}, \quad (1)$$

in $B_1 \subseteq \mathbb{R}^n$. The challenge is to understand the free boundary $\Gamma(u) := \partial\{u > 0\}$. L. Caffarelli showed (see [1]) that $u \in C_{\text{loc}}^{1,1}(B_1)$. Since we are only interested in the local structure of $\Gamma(u)$, we may assume $u \in C^{1,1}(\overline{B_1})$ with $\|u\|_{C^{1,1}(\overline{B_1})} \leq C_0$.

The free boundary $\Gamma(u)$ can be decomposed into the regular part $\text{Reg}(u)$ and the singular part $\Sigma(u)$, according to whether $r^{-2}u(x_0 + rx)$ converges to $\frac{1}{2}(e \cdot x)_+^2$ for some unit vector $e \in \mathbb{S}^{n-1}$ or a non-negative 2-homogeneous harmonic polynomial. Furthermore, $\text{Reg}(u)$ is relatively open in $\Gamma(u)$, and locally a $(n-1)$ -dimensional C^∞ (in fact, analytic) surface. In contrast, the structure of $\Sigma(u)$ is much more delicate, see [6] for various examples.

If $x_0 \in \Sigma(u)$, then there exists a unique non-negative 2-homogeneous polynomial p_{2,x_0} with $\Delta p_{2,x_0} = 1$, such that as $|x| \rightarrow 0$, we have

$$u(x_0 + x) = p_{2,x_0}(x) + |x|^2 \omega(|x|),$$

where ω is a modulus of continuity. Since $\{p_{2,x_0} = 0\}$ is a linear subspace of $\mathbb{R}^n$, whose dimension belongs to $\{0, 1, \dots, n-1\}$, we can give $\Sigma(u)$ the natural stratification as

$$\Sigma(u) = \bigcup_{m=0}^{n-1} \Sigma_m(u),$$

where $\Sigma_m(u)$ consists of all $x_0 \in \Sigma(u)$ such that $\dim\{p_{2,x_0} = 0\} = m$.

L. Caffarelli showed that $\Sigma_m(u)$ is locally contained in a m -dimensional C^1 surface. A natural question is whether $\Sigma_m(u)$ possesses higher regularity. Such results can be developed with the help of Whitney's extension theorem, as long as one obtain polynomial expansions of u up to some order.

Recall that if $x_0 \in \Sigma_0$, then u is strictly positive in a neighborhood of x_0 .¹ Hence we are left to consider Σ_m , where $m = 1, \dots, n-1$. In [3], Figalli et al. studied in detail the second order blow-up, and proved the following theorem:

Theorem 1.1. *Let u be the solution of the obstacle problem (1).*

- (1) *The top strata Σ_{n-1} is locally contained in a C^{1,α_0} manifold of dimension $(n-1)$, for some dimensional constant α_0 . Moreover, we can decompose $\Sigma_{n-1} := \Sigma_{n-1}^g \cup \Sigma_{n-1}^a$, with Σ_{n-1}^g locally contained in a $C^{1,1}$ manifold of dimension $(n-1)$, and Σ_{n-1}^a relatively open in Σ_{n-1} satisfying $\dim_{\mathcal{H}}(\Sigma_{n-1}^a) \leq n-3$ (and discrete if $n=3$).*

Moreover, if $n=2$, then Σ_1 is locally contained in a $C^{1,1}$ curve.

¹If not, suppose $0 \in \Sigma_0$, $0 \neq x_l \rightarrow 0$ and $u(x_l) = 0$. Let $r_l = |x_l|$ and $y_l = x_l/r_l$, then $y_l \rightarrow y_\infty \neq 0$. Let $r_l^{-2}u(r_l \cdot) \rightarrow p_2$, we have $p_2(y_l) = r_l^{-2}u(x_l) + o(1) = o(1)$, hence $p_2(y_\infty) = 0$, contradictory to $\{p_2 = 0\} = \{0\}$.

- (2) The lower strata $\Sigma_m (1 \leq m \leq n-2)$ is locally contained in a $C^{1, \log \varepsilon_0}$ manifold of dimension m , for some dimensional constant ε_0 . Moreover, we can decompose $\Sigma_m := \Sigma_m^g \cup \Sigma_m^a$, with Σ_m^g locally contained in a $C^{1,1}$ manifold of dimension m , and Σ_m^a relatively open in Σ_m satisfying $\dim_{\mathcal{H}}(\Sigma_m^a) \leq m-1$ (and Σ_1^a is discrete).

We remark that in fact [3] proved that when $n=2$, Σ_1 is locally contained in a C^2 curve. But we don't want to prove this, for it's about another branch of this theory that won't come up later anymore.

After that, they moved on and investigated blow-ups of higher order in [2]. Now new difficulty arose that the original monotonicity formulae fails. Their new method is complicated, and the blow-up procedure (or polynomial expansion) stops at order $5 - \zeta (\zeta > 0)$. Another new feature of this paper is that they proved the generic regularity (or Schaeffer's conjecture) up to dimension 4, which was only known to hold in $\mathbb{R}^2$ by Monneau (see [8]). To state this result, we consider a family of solutions $u(x, t): \overline{B_1} \times (-1, 1) \rightarrow \mathbb{R}$ (which means for each $t \in (-1, 1)$, $u(\cdot, t)$ is a solution to the obstacle problem (1)), with the uniform monotonicity property:

Definition 1.2. A family of solution $u(x, t): \overline{B_1} \times (-1, 1) \rightarrow \mathbb{R}$ is called uniformly monotone, if for every $t \in (-1, 1)$ and any compact set $K \subseteq \partial B_1 \cap \{u(\cdot, t) > 0\}$, there exists $c_{K_t} > 0$ such that

$$\inf_{x \in K_t} (u(\cdot, t') - u(\cdot, t)) \geq c_{K_t}(t' - t), \quad \forall t < t' < 1.$$

Also, we say $u(x, t)$ is a continuous family, if $u \in C(\overline{B_1} \times (-1, 1))$. Using the maximal principle, it's easy to see this continuity is equivalent to $u \in C(\partial B_1 \times (-1, 1))$. In [8], Monneau proved for arbitrary bounded smooth domain Ω , and set $t \in (0, \infty)$ and $u(\cdot, t) = t$ on $\partial\Omega$, that $\Sigma_t := \Sigma(u(\cdot, t)) = \emptyset$ for almost every t . The main result of [2] is the following. They also pointed out that their method will not imply generic regularity for higher dimension, even if one can prove that the polynomial expansion continues to arbitrary order.

Theorem 1.3. Suppose $u(x, t): \overline{B_1} \times (-1, 1)$ is a uniformly monotone and continuous family of solutions, then $\mathcal{H}^{n-4}(\Sigma_t) = 0$ for almost every $t \in (-1, 1)$.

In particular, if $n \leq 4$, then Σ_t is empty for almost every t .

It's still interesting that where one can prove a polynomial expansion to arbitrary order. To do this, the key issue is to prove a Lipschitz estimate to higher order blow-ups. [4] came up with a new approach, and ended this question with an affirmative answer. Intuitively, let $\Sigma^{kth} \subseteq \Sigma(u)$ be the set of points with polynomial expansion up to order k (later we'll give a precise definition), and $\Sigma^\infty := \cup_{k \geq 2} \Sigma^{kth}$ is the set of points which expands to ∞ .

Theorem 1.4. Let u be the solution of the obstacle problem (1), then $\dim_{\mathcal{H}}(\Sigma(u) \setminus \Sigma^\infty) \leq n-2$. Moreover, Σ^∞ is closed set locally contained in a $(n-1)$ -dimensional C^∞ manifold.

This article is organized as follows. The first two sections are contained in [3]. In Section 2 we introduce various monotonicity formulae, especially the Almgren frequency function, making preparations for the study of the second order blow-up in Section 3. The proof of Theorem 1.1 is also contained in Section 3. Then the next three sections studies higher order blow-ups as in [4]. Section 4 is devoted to the proof of the Lipschitz estimate, which is involved and technical. Section 5 introduces the truncated frequency function and studies higher order blow-ups. Section 6 makes refined estimates on the size of $\Sigma^{kth} \setminus \Sigma^{(k+1)th}$ and proves Theorem 1.4. Finally, the last

section deals with the generic regularity and proves Theorem 1.3. Many lemmas in Section 7 are similar to those in former sections, and their proofs are also similar (maybe a little more involved due to the appearance of the t variable). This article is self-contained, except for the theory of the thin obstacle problem (to be explained later). We barely list all the results on the homogeneous solution which turn out to be useful in Lemma 3.1.

2 Monotonicity Formulae

It's known that for harmonic functions, the Almgren frequency formula is monotone, and turns out to be useful in studying the properties of harmonic functions. Denote with P the set of all polynomials $p = \frac{1}{2}x^T Bx$, where B is a symmetric $n \times n$ non-negative matrix with $\text{tr}(B) = 1$. In other words, p is a non-negative 2-homogeneous polynomial with $\Delta p = 1$. We shall prove that the frequency function continues to be monotone for $u - p$ and $p \in P$. We point out here that p need not be the blow-up limit of u (which we denote with p_2). And we always assume $u \neq p (\forall p \in P)$ in any neighborhood of 0, for otherwise $\Sigma(u)$ is simple.

2.1 Almgren frequency function

We assume in this section that $w \in C_{\text{loc}}^{1,1}(B_1)$ and $r \in (0, 1)$. We define $w_r(x) := w(rx)$,

$$D(r, w) := r^{2-n} \int_{B_r} |\nabla w|^2 = \int_{B_1} |\nabla w_r|^2 \quad \text{and} \quad H(r, w) = r^{1-n} \int_{\partial B_r} w^2 = \int_{\partial B_1} w_r^2.$$

Let's compute their derivatives, which will be useful when proving various monotonicity formulae.

Lemma 2.1. *The derivatives of $D(r, w)$ and $H(r, w)$ are*

$$\begin{aligned} \frac{d}{dr} D(r, w) &= 2r^{-n} \int_{\partial B_r} |x \cdot \nabla w|^2 - 2r^{1-n} \int_{B_r} (x \cdot \nabla w) \Delta w, \\ \frac{d}{dr} H(r, w) &= 2r^{-n} \int_{\partial B_r} w(x \cdot \nabla w) = 2r^{1-n} \int_{B_r} (w \Delta w + |\nabla w|^2). \end{aligned}$$

Proof. By direct computation,

$$\begin{aligned}
\frac{d}{dr}D(r, w) &= \frac{d}{dr} \int_{B_1} r^2 |\nabla w(rx)|^2 = \frac{2}{r}D(r, w) + 2r^2 \sum_{i,j=1}^n \int_{B_1} x_i \partial_{ij} w(rx) \partial_j w(rx) \\
&= \frac{2}{r}D(r, w) + 2r^2 \sum_{i,j=1}^n \int_{B_1} x_i \partial_{ij} w(rx) \partial_j w(rx) \\
&= \frac{2}{r}D(r, w) + 2r^{1-n} \sum_{i,j=1}^n \int_{B_r} x_i \partial_{ij} w \partial_j w \\
&= \frac{2}{r}D(r, w) + 2r^{1-n} \sum_{i,j=1}^n \int_{B_r} \partial_j (x_i \partial_i w \partial_j w) - \partial_i w \partial_j (x_i \partial_j w) \\
&= \frac{2}{r}D(r, w) + 2r^{1-n} \sum_{i,j=1}^n \int_{\partial B_r} \frac{x_j}{r} \cdot x_i \partial_i w \partial_j w - 2r^{1-n} \int_{B_r} (|\nabla w|^2 + (x \cdot \nabla w) \Delta w) \\
&= 2r^{-n} \int_{\partial B_r} |x \cdot \nabla w|^2 - 2r^{1-n} \int_{B_r} (x \cdot \nabla w) \Delta w, \\
\frac{d}{dr}H(r, w) &= \frac{d}{dr} \int_{\partial B_1} w(rx)^2 = 2 \int_{\partial B_1} w(rx) (x \cdot \nabla w(rx)) = 2r^{-n} \int_{\partial B_r} w(x \cdot \nabla w) \\
&= 2r^{1-n} \int_{B_r} (w \Delta w + |\nabla w|^2). \quad \square
\end{aligned}$$

The Almgren frequency function is

$$\phi(r, w) := \frac{D(r, w)}{H(r, w)} = \frac{r^{2-n} \int_{B_r} |\nabla w|^2}{r^{1-n} \int_{\partial B_r} w^2}.$$

Since we are assuming $u \neq p$ in any neighborhood of 0, the denominator never vanishes.

It's known that if w is a harmonic function, then $\phi(r, w)$ is monotone increasing in r . If $w := u - p$, where u is the solution of obstacle problem (1) and $p \in P$, we shall prove that $\phi(r, w)$ is also monotone increasing. To this end, we need auxillary lemmas on the Weiss function, which is defined as

$$W(r, w) := r^{-2-n} \int_{B_r} (|\nabla w|^2 + 2u) - 2r^{-3-n} \int_{\partial B_r} u^2.$$

Lemma 2.2. *If $0 \in \Sigma(u)$, then $\frac{d}{dr}W(r, u) \geq 0$. Moreover, for any $p \in P$ and $r > 0$, we have*

$$W(0+, u) = \frac{\omega_n}{2(n+2)} = W(r, p),$$

where ω_n is the volumn of the unit ball in $\mathbb{R}^n$.

Proof. By Lemma 2.1, since $\Delta u = \chi_{\{u>0\}}$ and $\nabla u = 0$ in $\{u = 0\}$,

$$\begin{aligned}
\frac{d}{dr}D(r, u) &= 2r^{-n} \int_{\partial B_r} |x \cdot \nabla u|^2 - 2r^{1-n} \int_{B_r} x \cdot \nabla u, \\
\frac{d}{dr}H(r, u) &= 2r^{-n} \int_{\partial B_r} u(x \cdot \nabla u) = 2r^{1-n} \int_{B_r} u + 2r^{-1}D(r, u).
\end{aligned}$$

Therefore

$$\begin{aligned}
\frac{d}{dr}W(r, u) &= \frac{d}{dr} \left(r^{-4}(D(r, u) - 2H(r, u)) + 2r^{-2} \int_{B_1} u(rx) \right) \\
&= -4r^{-5}(D(r, u) - 2H(r, u)) + r^{-4} \frac{d}{dr}(D(r, u) - 2H(r, u)) - 4r^{-3-n} \int_{B_r} u + 2r^{-3-n} \int_{B_r} x \cdot \nabla u \\
&= -4r^{-5}D(r, u) + 8r^{-5}H(r, u) + 2r^{-4-n} \int_{\partial B_r} |x \cdot \nabla u|^2 - 2r^{-4} \frac{d}{dr}H(r, u) - 4r^{-3-n} \int_{B_r} u \\
&= 8r^{-4-n} \int_{\partial B_r} u^2 + 2r^{-4-n} \int_{\partial B_r} |x \cdot \nabla u|^2 - 8r^{-4-n} \int_{\partial B_r} u(x \cdot \nabla u) \\
&= 2r^{-4-n} \int_{\partial B_r} (2u - (x \cdot \nabla u))^2 \geq 0
\end{aligned}$$

For any $p \in P$, note that $\Delta p = 1$ and $x \cdot \nabla p = p$ (since p is 2-homogeneous),

$$\begin{aligned}
W(r, p) &= r^{-2-n} \int_{B_r} (|\nabla p|^2 + p\Delta p + p) - r^{-3-n} \int_{\partial B_r} p(x \cdot \nabla p) \\
&= r^{-2-n} \int_{B_r} (\operatorname{div}(p\nabla p) + p) - r^{-2-n} \int_{B_r} \operatorname{div}(p\nabla p) \\
&= r^{-2-n} \int_{B_r} p = \int_{B_1} p.
\end{aligned}$$

After a change of coordinates, $p = \frac{1}{2}(a_1x_1^2 + \dots + a_nx_n^2)$ for some $a_i \geq 0$ and $\sum_i a_i = 1$. One easily calculates for $i = 1, \dots, n$,

$$\int_{B_1} x_i^2 = \frac{1}{n} \int_{B_1} |x|^2 = \frac{1}{n} \int_0^1 \mathcal{H}^{n-1}(S^{n-1})r^{n+1} dr = \frac{\omega_n}{n+2}.$$

This implies $W(r, p) = \frac{\omega_n}{2(n+2)}$. Finally, since $0 \in \Sigma(u)$, we have $u(x) = p_2(x) + o(|x|^2)$ for some $p_2 \in P$, then it's easy to verify $W(0+, u) = W(0+, p) = W(r, p)$. $\square$

Corollary 2.3. *If $0 \in \Sigma(u)$, $p \in P$ and $w := u - p$, then $\phi(r, w) \geq 2$, and*

$$r^{1-n} \int_{\partial B_r} w(x \cdot \nabla w - 2w) \geq r^{2-n} \int_{B_r} w\Delta w \geq 0.$$

Proof. For any $r > 0$, since $W(r, u) \geq W(0+, u) = W(r, p)$ and $\Delta p = 1$,

$$\begin{aligned}
0 \leq W(r, u) - W(r, p) &= r^{-2-n} \int_{B_r} (|\nabla w|^2 + 2\nabla w \cdot \nabla p + 2w) - 2r^{-3-n} \int_{\partial B_r} (w^2 + 2wp) \\
&= r^{-2-n} \int_{B_r} (|\nabla w|^2 + 2\operatorname{div}(w\nabla p)) - 2r^{-3-n} \int_{\partial B_r} (w^2 + 2wp) \\
&= r^{-4}(D(r, w) - 2H(r, w)) + 2r^{-3-n} \int_{\partial B_r} w(x \cdot \nabla p - 2p) = r^{-4}(D(r, w) - 2H(r, w)),
\end{aligned}$$

where we used p is 2-homogeneous, hence $x \cdot \nabla p = 2p$. This implies $\phi(r, w) \geq 2$.

Finally, the last inequality follows from $D(r, w) \geq 2H(r, w)$,

$$D(r, w) = r^{2-n} \int_{B_r} \operatorname{div}(w \nabla w) - w \Delta w = r^{1-n} \int_{\partial B_r} w(x \cdot \nabla w) - r^{2-n} \int_{B_r} w \Delta w.$$

and $w \Delta w = w(\chi_{\{u>0\}} - 1) = p\chi_{\{u=0\}} \geq 0$. □

Now we are in the position to prove $\phi(r, w)$ is monotone increasing for $w = u - p$.

Lemma 2.4. *If $0 \in \Sigma(u)$, $p \in P$ and $w := u - p$, then*

$$\frac{d}{dr} \log \phi(r, w) \geq \frac{2}{r} \frac{\left(r^{2-n} \int_{B_r} w \Delta w \right)^2}{D(r, w) H(r, w)} \geq 0$$

Proof. By Lemma 2.1 and $(x \cdot \nabla w) \Delta w = 2p\chi_{\{u=0\}} = 2w \Delta w$.

$$\begin{aligned} \frac{d}{dr} D(r, w) &= 2r^{-n} \int_{\partial B_r} |x \cdot \nabla w|^2 - 4r^{1-n} \int_{B_r} w \Delta w, \\ \frac{d}{dr} H(r, w) &= 2r^{-n} \int_{\partial B_r} w(x \cdot \nabla w) = 2r^{1-n} \int_{B_r} w \Delta w + 2r^{-1} D(r, w). \end{aligned}$$

Since $\phi'/\phi = D'/D - H'/H$ (' refers to $\frac{d}{dr}$), we obtain

$$\begin{aligned} D(r, w) H(r, w) \frac{\phi'(r, w)}{\phi(r, w)} &= D'(r, w) H(r, w) - H'(r, w) D(r, w) \\ &= \left(2r^{-n} \int_{\partial B_r} |x \cdot \nabla w|^2 - 4r^{1-n} \int_{B_r} w \Delta w \right) \left(r^{1-n} \int_{\partial B_r} w^2 \right) \\ &\quad - \left(2r^{-n} \int_{\partial B_r} w(x \cdot \nabla w) \right) \left(r^{1-n} \int_{\partial B_r} w(x \cdot \nabla w) - r^{2-n} \int_{B_r} w \Delta w \right) \\ &= 2r^{1-2n} \left(\left(\int_{\partial B_r} |x \cdot \nabla w|^2 \right) \left(\int_{\partial B_r} w^2 \right) - \left(\int_{\partial B_r} w(x \cdot \nabla w) \right)^2 \right) \\ &\quad + 2 \left(r^{2-n} \int_{B_r} w \Delta w \right) \left(r^{-n} \int_{\partial B_r} w(x \cdot \nabla w - 2w) \right) \end{aligned}$$

The first term is non-negative by the Cauchy–Schwarz inequality. For the second term, use Corollary 2.3, and we are done. □

2.2 Frequency controls the growth of $H(r, w)$

If w is a harmonic function, it's known that for any $r < R$ we have

$$H(R, w) \leq (R/r)^{2\phi(R, w)} H(r, w).$$

In other words, the frequency function controls the order of growth of $H(r, w)$. We show that this is also true for $w = u - p$, where $0 \in \Sigma(u)$ and $p \in P$. We make one more definition that

$$H_\lambda(r, w) := r^{-2\lambda} H(r, w).$$

Lemma 2.5. *Let $0 \in \Sigma(u)$, $p \in P$, $w := u - p$, and $0 < r_1 < r_2 < 1$. We have:*

(1) *If $\phi(r, w) \geq \underline{\lambda} \geq 0$, then*

$$\frac{H(r_2, w)}{H(r_1, w)} \geq \left(\frac{r_2}{r_1}\right)^{2\underline{\lambda}}.$$

In other words, $H_{\underline{\lambda}}(r, w) = r^{-2\underline{\lambda}}H(r, w)$ is monotone increasing.

(2) *If $\phi(r_2, w) \leq \bar{\lambda}$, then for any $\delta > 0$, there exists C_δ depending on $n, \bar{\lambda}, \delta$, such that*

$$\frac{H(r_2, w)}{H(r_1, w)} \leq C_\delta \left(\frac{r_2}{r_1}\right)^{2\bar{\lambda} + \delta}.$$

Proof. (1) By Lemma 2.1 and $w\Delta w = p\chi_{\{u=0\}} \geq 0$, we obtain $H'/H \geq 2r^{-1}\phi$, thus

$$\frac{H'_{\underline{\lambda}}(r, w)}{H_{\underline{\lambda}}(r, w)} = \frac{H'}{H} - \frac{2\underline{\lambda}}{r} \geq \frac{2}{r}(\phi(r, w) - \underline{\lambda}) \geq 0,$$

for any $r \in [r_1, r_2]$, since $\phi(r, w)$ is monotone increasing.

(2) Let

$$F(r, w) := \frac{r^{2-n} \int_{B_r} w \Delta w}{H(r, w)} \geq 0.$$

Then $H'/H \geq 2r^{-1}(\phi + F)$.

Lemma 2.4 says $\phi' \geq 2r^{-1}F^2$. Integrate this and apply the Cauchy–Schwartz inequality,

$$\begin{aligned} \left(\int_{r_1}^{r_2} r^{-1} F(r, w) dr \right)^2 &\leq \left(\int_{r_1}^{r_2} r^{-1} (F(r, w))^2 dr \right) \left(\int_{r_1}^{r_2} r^{-1} dr \right) \\ &\leq C \log(r_2/r_1) \left(\int_{r_1}^{r_2} \phi'(r, w) dr \right)^{1/2} \leq C \log(r_2/r_1), \end{aligned}$$

where C depends on $n, \bar{\lambda}$. Therefore

$$\begin{aligned} \log \frac{H(r_2, w)}{H(r_1, w)} &\leq \int_{r_1}^{r_2} 2r^{-1}(\bar{\lambda} + F(r, w)) dr \\ &\leq 2\bar{\lambda} \log(r_2/r_1) + C(\log(r_2/r_1))^{1/2} \leq (2\bar{\lambda} + \delta) \log(r_2/r_1) + C_\delta, \end{aligned}$$

where C_δ depends on $n, \bar{\lambda}, \delta$. □

Remark 2.6. *Due to the monotonicity of $\phi(r, w)$, for any $\lambda \leq \phi(0+, w)$, we obtain that $H_\lambda(r, w)$ is monotone increasing provided . In particular, Corollary 2.3 implies $H_2(r, w)$ is always increasing.*

Remark 2.7. *This lemma implies that, for any $\delta > 0$, as $r \rightarrow 0$,*

$$r^{2\phi(0+, u-p)+\delta} \lesssim H(r, u-p) \lesssim r^{2\phi(0+, u-p)}.$$

As a corollary, if $\lambda > \phi(0+, u-p)$, then $H_\lambda(r, u-p) = r^{-2\lambda}H(r, u-p) \rightarrow \infty$.

2.3 Truncated frequency function

The Almgren frequency function behaves well for $w = u - p$. But it's not longer monotone (or even “almost” monotone) when studying higher order blow-ups. The truncated version will come into use, namely for $\gamma > 0$ we define

$$\phi^\gamma(r, w) := \frac{D(r, w) + \gamma r^{2\gamma}}{H(r, w) + r^{2\gamma}}.$$

If $0 < r < 1$, then as $\gamma \rightarrow \infty$, the truncated frequency function converges to the usual Almgren frequency function. Lemma 5.10 will justify the name “truncated frequency”.

We only calculate its derivative in this subsection, and leave the proof of almost monotonicity in Lemma 5.2 after proving Lipschitz estimates Lemma 3.8.

Lemma 2.8. *For all $w \in C_{loc}^{1,1}(B_1)$, $r \in (0, 1)$ and $\gamma > 0$, we have*

$$\frac{d}{dr} \phi^\gamma(r, w) \geq \frac{2}{r} \frac{(r^{2-n} \int_{B_r} w \Delta w)^2 + E^\gamma(r, w)}{(H(r, w) + r^{2\gamma})^2},$$

where

$$E^\gamma(r, w) := \left(r^{2-n} \int_{B_r} w \Delta w \right) (D(r, w) + \gamma r^{2\gamma}) - \left(r^{2-n} \int_{B_r} (x \cdot \nabla w) \Delta w \right) (H(r, w) + r^{2\gamma}).$$

In particular, we have

$$\frac{d}{dr} \phi^\gamma(r, w) \geq \frac{2}{r} \frac{\int_{B_1} (\phi^\gamma(r, w) w_r - x \cdot \nabla w_r) \Delta w_r}{H(r, w) + r^{2\gamma}}.$$

Proof. By Lemma 2.1 (we drop w to make notations simpler),

$$\begin{aligned} \frac{r}{2} (H(r) + r^{2\gamma})^2 \frac{d}{dr} \phi^\gamma(r) &= \frac{r}{2} \left((H(r) + r^{2\gamma}) \frac{d}{dr} (D(r) + \gamma r^{2\gamma}) - (D(r) + \gamma r^{2\gamma}) \frac{d}{dr} (H(r) + \gamma r^{2\gamma}) \right) \\ &= (H(r) + r^{2\gamma}) \left(\int_{\partial B_1} |x \cdot \nabla w_r|^2 - \int_{B_1} (x \cdot \nabla w_r) \Delta w_r + \gamma^2 r^{2\gamma} \right) \\ &\quad - (D(r) + \gamma r^{2\gamma}) \left(\int_{\partial B_1} w_r (x \cdot \nabla w_r) + \gamma r^{2\gamma} \right) \\ &= \left(\int_{\partial B_1} |x \cdot \nabla w_r|^2 + \gamma^2 r^{2\gamma} \right) \left(\int_{\partial B_1} w_r^2 + r^{2\gamma} \right) \\ &\quad - \left(\int_{\partial B_1} w_r (x \cdot \nabla w_r) + \gamma r^{2\gamma} \right)^2 + R(r, w) \\ &\geq R^\gamma(r, w), \end{aligned}$$

where we used the Cauchy-Schwarz inequality, and $R^\gamma(r, w)$ represents

$$\begin{aligned}
R^\gamma(r, w) &= -(H(r, w) + r^{2\gamma}) \int_{B_1} \partial_n w_r \Delta w_r + \int_{B_1} w_r \Delta w_r \left(\int_{\partial B_1} w_r \partial_n w_r + \gamma r^{2\gamma} \right) \\
&= -(H(r) + r^{2\gamma}) \int_{B_1} \partial_n w_r \Delta w_r + \int_{B_1} w_r \Delta w_r \left(\int_{B_1} (|\nabla w_r|^2 + w_r \Delta w_r) + \gamma r^{2\gamma} \right) \\
&= \left(\int_{B_r} w_r \Delta w_r \right)^2 + E^\gamma(r, w). \quad \square
\end{aligned}$$

If we can show the “error term” $E^\gamma(r, w) \geq -Cr^{\varepsilon-1}$, then we obtain the required almost monotonicity formula, and $\phi^\gamma(0+, w)$ exists. However, this estimate doesn’t hold in general and is highly non-trivial in our specific case $w := u - P_k$ (to be defined later).

To end this section, we show that under the almost monotonicity assumption, ϕ^γ controls the growth $H(r, w) + r^{2\gamma}$, which is similar to Lemma 2.5.

Lemma 2.9. *Let $w \in C_{loc}^{1,1}(B_1)$, and $0 < r_1 < r_2 < 1$. Suppose*

$$\frac{d}{dr} \phi^\gamma(r, w) \geq \frac{2}{r} \frac{\left(r^{2-n} \int_{B_r} w \Delta w \right)^2}{(H(r, w) + r^{2\gamma})^2} - C_0 r^{\varepsilon-1},$$

For any $\delta > 0$, we have:

(1) *If $0 \leq \underline{\lambda} \leq \phi^\gamma(r, w) \leq \bar{\lambda} \geq 0$ for $r \in [r_1, r_2]$, then*

$$c_\delta \left(\frac{r_2}{r_1} \right)^{2\bar{\lambda}-\delta} \leq \frac{H(r_2, w) + r_2^{2\gamma}}{H(r_1, w) + r_1^{2\gamma}} \leq C_\delta \left(\frac{r_2}{r_1} \right)^{2\bar{\lambda}+\delta},$$

where c_δ, C_δ depends on $n, \gamma, \varepsilon, C_0, \bar{\lambda}, \delta$.

(2) *Assume in addition that*

$$\frac{r^{2-n} \int_{B_r} w \Delta w}{H(r, w) + r^{2\gamma}} \geq -C_0 r^\varepsilon.$$

Then we have $\phi^\gamma(0+, w) \leq \gamma$, and

$$\frac{H(r_2, w) + r_2^{2\gamma}}{H(r_1, w) + r_1^{2\gamma}} \geq c \left(\frac{r_2}{r_1} \right)^{2\phi^\gamma(0+, w)},$$

where c depends on $n, \gamma, C_0, \varepsilon$.

Proof. (1) Let’s define (it’s different from the one in the proof of Lemma 2.5, maybe F^γ is better)

$$F(r) := \frac{r^{2-n} \int_{B_r} w \Delta w}{H(r, w) + r^{2\gamma}}.$$

Again by Lemma 2.1 one easily obtain

$$\frac{d}{dr} \log(H(r, w) + r^{2\gamma}) = \frac{2}{r} \phi^\gamma(r, w) + \frac{2}{r} F(r).$$

We need to estimate $F(r)$. The assumption says $(\phi^\gamma(r))' + C_0 r^{\varepsilon-1} \geq \frac{2}{r}(F(r))^2$, so

$$\begin{aligned} \left(\int_{r_1}^{r_2} \frac{1}{r} F(r) \right)^2 &\leq \left(\int_{r_1}^{r_2} \frac{1}{r} (F(r))^2 \right) \left(\int_{r_1}^{r_2} \frac{1}{r} \right) \\ &\leq C \log(r_2/r_1) \left(\int_{r_1}^{r_2} (\phi^\gamma(r))' + r^{\varepsilon-1} \right) \\ &\leq C \log(r_2/r_1) (\bar{\lambda} + r_2^\varepsilon) \leq C \log(r_2/r_1). \end{aligned}$$

Therefore

$$\begin{aligned} \log \frac{H(r_2) + r_2^{2\gamma}}{H(r_1) + r_1^{2\gamma}} &\leq \int_{r_1}^{r_2} \frac{2}{r} (\phi^\gamma(r, w) + F(r)) \\ &\leq C(\log(r_2/r_1))^{1/2} + 2\bar{\lambda} \log(r_2/r_1) \\ &\leq (2\bar{\lambda} + \delta) \log(r_2/r_1) + C_\delta. \end{aligned}$$

The other half of the inequality follows in the same ways, using $\phi^\gamma \geq \underline{\lambda}$.

(2) In this case $F(r) \geq -r^\varepsilon$. We also have

$$\phi^\gamma(r) + C_0 r^\varepsilon \geq \phi^\gamma(0+),$$

due to the monotonicity. Therefore

$$\begin{aligned} \log \frac{H(r_2) + r_2^{2\gamma}}{H(r_1) + r_1^{2\gamma}} &\geq \int_{r_1}^{r_2} \frac{2}{r} (\phi^\gamma(r, w) + F(r)) \\ &\geq \int_{r_1}^{r_2} \frac{2}{r} (\phi^\gamma(0+) - C r^\varepsilon) \\ &= 2\phi^\gamma(0+) \log(r_2/r_1) - C r^\varepsilon \\ &\geq 2\phi^\gamma(0+) \log(r_2/r_1) - C. \end{aligned}$$

Finally, if $\phi^\gamma(0+) > \gamma$, then we can pick $\gamma < \underline{\lambda} < \phi^\gamma(0+) < \bar{\lambda}$, and we have $\underline{\lambda} \leq \phi^\gamma(r) \leq \bar{\lambda}$ for $r \ll 1$. By (1) we obtain $H(r) + r^{2\gamma} \lesssim r^{2\underline{\lambda}-\delta}$. But $2\gamma \leq 2\underline{\lambda} - \delta$ for $\delta \ll 1$. Contradictory. $\square$

3 Second order blow-up

In this section, we apply the monotonicity formula develop above to study the second order blow-up

$$\tilde{w}_r := \frac{u_r - p_2}{\|(u - p_2)_r\|_{L^2(\partial B_1)}},$$

where $(u - p_2)_r := (u - p_2)(r \cdot)$, $0 \in \Sigma(u)$ and $r^{-2}u_r \rightarrow p_2$. By the Almgren monotonicity formula, $\|\nabla \tilde{w}_r\|_{L^2(B_1)}$ is increasing, and since $\|\tilde{w}_r\|_{L^2(\partial B_1)} = 1$, as $r \rightarrow 0$ these $\tilde{w}_r$ is bounded in $H^1(B_1)$ by the Poincaré's inequality. Therefore, it contains a subsequence converging to some q in weak H^1 . Since $u_r(x) = r^2 p_2(x) + o(r^2)$ in B_1 , we expect that $\{q = 0\}$ is almost $\{p_2 = 0\}$. Then a difference appears for the cases $\dim\{p_2 = 0\} \leq n - 2$ and $\dim\{p_2 = 0\} = n - 1$. In the former case, q is harmonic. However, in the latter case, q need not be harmonic, and it turns out to be a solution of the thin obstacle problem (or the Signorini problem).

3.1 The thin obstacle problem

In this subsection we gather together known results that will come into use on the thin obstacle problem. “Thin obstacle” means the obstacle is in a $(n - 1)$ -dimensional subset, rather than the whole domain. More precisely, a solution to the thin obstacle problem (with obstacle 0 on L) means

$$\begin{cases} \Delta q \leq 0 \text{ and } q\Delta q = 0 & \text{in } \mathbb{R}^n, \\ \Delta q = 0 & \text{in } \mathbb{R}^n \setminus L, \\ q \geq 0 & \text{on } L, \end{cases} \quad (2)$$

where $L \subseteq \mathbb{R}^n$ is a $(n - 1)$ -dimensional subspace. It’s shown by Caffarelli that the solution q has optimal regularity $C_{\text{loc}}^{1,1/2}$.

Our interest is in homogeneous solutions. For $\lambda \geq 0$ and hyperplane $L \subseteq \mathbb{R}^n$, let $S_\lambda(L)$ consist of all λ -homogeneous solution of thin obstacle equation (2). For some $v \in S^{n-1}$, we have $L = v^\perp$, and each solution q can be split into the even part and odd part:

$$q^{\text{even}}(x) := \frac{q(x) + q(x - 2(x \cdot v)v)}{2}, \quad q^{\text{odd}} := \frac{q(x) - q(x - 2(x \cdot v)v)}{2}.$$

Clearly q^{even} and q^{odd} still solve (2). It’s then natural to define $S_\lambda^{\text{even}}(L)$ and $S_\lambda^{\text{odd}}(L)$, whose meaning needs no explanation.

For each solution q , the singular set is defined as

$$\Sigma(q) := \{x \in L \mid q = |\nabla q| = 0\}.$$

If q is even with respect to L , then $\Sigma(q) = L \cap \{q = 0\}$.

It’s easy to see that, if q is a solution to the thin obstacle problem, then q^{odd} is in fact a harmonic function. One should differ the definition above from the singular set of harmonic functions. That is, if w is a harmonic function, then $\Sigma(w) := \{w = |\nabla w| = 0\}$. We will unfortunately encounter these two definitions in one proof, we’ll specify which meaning we are using at that time.

The Almgren frequency function is monotone increasing for q (the proof is the same as for harmonic functions, since it only requires $q\Delta q = 0$). Therefore we may refer to $\phi(0+, q)$ as the frequency of q . And, $\phi(r, q) \equiv \lambda$ if and only if q is λ -homogeneous.

We list here what we shall need in the future without proof. For a proof of these results, see e.g. [2, 3, 4] as well as references therein.

Lemma 3.1. (1) For $m \in \mathbb{N}$, S_{2m+1} vanishes on L .

(2) $S_\lambda^{\text{odd}}(L)$ consists exactly of those λ -homogeneous harmonic polynomials vanishing on L , thus it’s empty if $\lambda \notin \mathbb{N}$.

(3) For $m \in \mathbb{N}$, S_{2m}^{even} consists exactly of those $(2m)$ -homogeneous harmonic polynomials that are non-negative on L .

(4) If $m \in \mathbb{N}$ and $q \in S_{2m+1}^{\text{even}}(x_n = 0)$, then

$$q(x) = -|x_n|(q_0(x) + x_n^2 q_1(x)),$$

where q_0, q_1 are polynomials with $q_0 \geq 0$.

(5) For $n = 2$, all possible homogeneity λ consist of

$$\mathbb{N} \cup \{2m - 1/2 \mid m \in \mathbb{N}\}.$$

(6) For $n \geq 2$, we have

$$\phi(0+, q(x + \cdot)) \in \mathbb{N} \cup \{2m - 1/2 \mid m \in \mathbb{N}\}$$

for $\mathcal{H}^{n-3}$ -a.e. $x \in L \cap \{q = 0\}$.

3.2 Blow-up analysis

The target of this subsection is the following proposition. Recall that $\tilde{w}_r := (u - p_2)_r / H(r, u - p_2)^{1/2}$, where $0 \in \Sigma(u)$, $r^{-2}u_r \rightarrow p_2$.

Proposition 3.2. *Let $L := \{p_2 = 0\}$, $m = \dim L \in \{0, 1, \dots, n - 1\}$ and $\lambda_2 := \phi(0+, w)$, then:*

1. *For $0 \leq m \leq n - 2$, we have $\lambda_2 \in \{2, 3, \dots\}$. Moreover, for every sequence $r_k \rightarrow 0$, up to a subsequence $\tilde{w}_{r_k} \rightharpoonup q$ in $H_{loc}^1(\mathbb{R}^n)$, where $q \neq 0$ is a λ_2 -homogeneous harmonic polynomial. In addition, if $\lambda_2 = 2$, then after a suitable change of coordinates,*

$$\nabla^2 p_2 = \left(\begin{array}{ccc|ccc} \mu_1 & & & & & \\ & \ddots & & & & \\ & & \mu_{n-m} & & & \\ \hline & & & 0_{m \times (n-m)} & & \\ 0_{m \times (n-m)} & & & & 0_{m \times m} & \end{array} \right) \quad \text{and} \quad \nabla^2 q = \left(\begin{array}{ccc|ccc} t & & & & & \\ & \ddots & & & & \\ & & t & & & \\ \hline & & & 0_{m \times (n-m)} & & \\ 0_{m \times (n-m)} & & & & -N & \end{array} \right), \quad (3)$$

where $\mu_1, \dots, \mu_{n-m}, t > 0$, and N is an $m \times m$ matrix with $\text{tr}(N) = (n - m)t$.

2. *For $m = n - 1$, we have $\lambda_* \geq 2 + \alpha_0$ for some dimensional constant $\alpha_0 > 0$. Moreover, for every sequence $r_k \rightarrow 0$, up to a subsequence $\tilde{w}_{r_k} \rightharpoonup q$ in $H_{loc}^1(\mathbb{R}^n)$ and $\tilde{w}_{r_k} \rightarrow q$ in $C_{loc}^0(\mathbb{R}^n)$, where $q \neq 0$ is a λ_2 -homogeneous solution of the thin obstacle problem (with obstacle 0 on L).*

We shall need some auxiliary lemmas.

Lemma 3.3. *Assume $\tilde{w}_{r_k} \rightharpoonup q$ in $H^1(B_1)$ as in Proposition 3.2, then for any $p \in P$,*

$$\int_{\partial B_1} q(p_2 - p) \geq 0. \quad (4)$$

Proof. Let $h_r := \|w_r\|_{L^2(\partial B_1)} = o(r^2)$ and $\varepsilon_r := h_r/r^2 = o(1)$. By the compactness of the trace operator, $\tilde{w}_{r_k} \rightarrow q$ in $L^2(\partial B_1)$. By Remark 2.6, we have

$$H_2(r, u - p) = \int_{\partial B_1} \left(\frac{w_r}{r^2} + p_2 - p \right)^2 \downarrow \int_{\partial B_1} (p_2 - p)^2.$$

Since $r^{-2}w_r = \varepsilon_r \tilde{w}_r$, substituting into the inequality above,

$$\varepsilon_{r_k}^2 \int_{\partial B_1} \tilde{w}_{r_k}^2 + 2\varepsilon_{r_k} \int_{\partial B_1} \tilde{w}_{r_k} (p_2 - p) \geq 0.$$

Dividing by ε_{r_k} and let $k \rightarrow \infty$, we are done. $\square$

Lemma 3.4. *Let q be a 2-homogeneous harmonic polynomial, such that (4) holds for some $p_2 \in P$, then after a suitable change of coordinates, (3) holds.*

Proof. Let $A = \nabla^2 p_2$, $B = \nabla^2 p$, $C = \nabla^2 q$. Then (4) says (note that $x \cdot \nabla q = 2q$ and $\Delta q = 0$)

$$0 \leq \int_{\partial B_1} q(p_2 - p) = \frac{1}{2} \int_{\partial B_1} (x \cdot \nabla q)(p_2 - p) = \frac{1}{2} \int_{B_1} \nabla q \cdot \nabla (p_2 - p) = \frac{1}{2} \int_{B_1} Cx \cdot (A - B)x.$$

It's obvious that $\int_{B_1} x^T M x = c_n \operatorname{tr}(M)$ for any symmetric matrix M , where c_n is a dimensional constant (cf. the proof of Lemma 2.2). Hence we obtain

$$\operatorname{tr}(CA) \geq \operatorname{tr}(CB),$$

for all non-negative $n \times n$ symmetric matrix B with $\operatorname{tr}(B) = 1$.

We show that this inequality implies (3). Let $e \in S^{n-1}$ be an eigenvector for C corresponding to largest eigenvalue $\nu_{\max} > 0$, and choose $B = e \otimes e$, then

$$\nu_{\max} = \operatorname{tr}(\nu_{\max} A) \geq \operatorname{tr}(CA) \geq \operatorname{tr}(CB) = \nu_{\max}.$$

Thus $\operatorname{tr}((\nu_{\max} \operatorname{Id} - C)A) = 0$. But $\nu_{\max} \operatorname{Id} - C$ and A are both non-negative. Set $L := \{p_2 = 0\} = \ker(A)$, we must have $(\nu_{\max} \operatorname{Id} - C)_{L^\perp} = 0$. Then one easily verifies (3). $\square$

Lemma 3.5. *Fix $R > 0$, then for $r < 1/R$, $\tilde{w}_r$ is bounded in $H^1(B_R)$. Moreover, the bound only depends on $n, R, \phi(1, u - p)$.*

Proof. We have shown at the beginning of this subsection that $\tilde{w}_r$ is bounded in $H^1(B_1)$. To prove for any $R > 0$, one simply applies Lemma 2.5 to see that, for $r < 1/R$ we have

$$R^{2\phi(0+, w)} \leq \frac{H(R, \tilde{w}_r)}{H(1, \tilde{w}_r)} = \frac{H(Rr, w)}{H(r, w)} \leq C_\delta R^{2\phi(1, w) + \delta},$$

for any $\delta > 0$. Both sides are just constants, so we are done by the same argument. $\square$

Lemma 3.6. *Let $0 \in \Sigma(u)$, $p \in P$ and $w = u - p$, then for $r < 1/2$,*

$$\|w_r\|_{L^\infty(B_1)} \leq C(n) \|w_r\|_{L^2(B_2 \setminus B_{1/2})}.$$

Proof. Since $\Delta w = -\chi_{\{u=0\}} \leq 0$, by mean-value inequality, for some $z \in \partial B_1$ we have

$$\min_{B_1} w_r = w_r(z) \geq \int_{B_{1/2}(z)} w_r \geq -C \|w_r\|_{L^2(B_2 \setminus B_{1/2})}.$$

To obtain upper bound, note that w is non-positive inside $\{u = 0\} \cap B_r$, so w_+ is superharmonic, and in the same manner one concludes

$$\max_{B_1} w_r \leq C \|w\|_{L^2(B_2 \setminus B_{1/2})}. \quad \square$$

Corollary 3.7. *For any fixed $R > 0$ and $r < 1/(2R)$, we have $\|\tilde{w}_r\|_{L^\infty(B_R)} \leq C(n, R, \phi(1, u - p))$.*

Proof. Apply Lemma 3.5 and Lemma 3.6. $\square$

Lemma 3.8. *For any fixed $R > 0$ and $r < 1/(2R)$, let $w = u - p$, where $0 \in \Sigma(u)$ and $p \in P$, then we have*

$$\partial_{ee}\tilde{w}_r \geq -C \text{ in } B_R, \quad \forall e \in S^{n-1} \cap \{p = 0\},$$

where C depends on $n, R, \phi(1, u - p)$.

In addition, if $\dim\{p = 0\} = n - 1$, then we obtain a Lipschitz estimate (where C also depends on $n, R, \phi(1, u - p)$)

$$|\nabla \tilde{w}_r| \leq C \text{ in } B_{R/2}.$$

Proof. For any function f , $e \in S^{n-1}$ and $h > 0$, define the second order incremental quotient

$$\delta_{e,h}^2 f := \frac{f(\cdot + he) + f(\cdot - he) - 2f(\cdot)}{h^2}.$$

In $B_R \cap \{u_r = 0\}$, $\delta_{e,h}^2 w_r \geq 0$ since $u \geq 0$ and $\delta_{e,h}^2 p \geq 0$. In $B_R \setminus \{u_r = 0\}$, since $\Delta u(r \cdot) = 1$

$$\Delta(\delta_{e,h}^2 w_r) = \frac{\Delta u_r(\cdot + he) + \Delta u_r(\cdot - he) - 2\Delta u_r(\cdot)}{h^2} \leq 0.$$

Therefore $(\delta_{e,h}^2 \tilde{w}_r)_-$ is superharmonic, so does its limit $(\partial_{ee}^2 \tilde{w}_r)_-$ (keeping in mind $u_r \in C^{1,1}(B_R)$). By weak Harnack inequality, for any $0 < \varepsilon < 1$,

$$\|(\partial_{ee}^2 \tilde{w}_r)_-\|_{L^\infty(B_R)} \leq C(n, \varepsilon) \left(\int_{B_{3R/2}} (\partial_{ee} \tilde{w}_r)_-^\varepsilon \right)^{1/\varepsilon} \leq C(n, \varepsilon) \left(\int_{B_{3R/2}} |\partial_{ee}^2 \tilde{w}_r|^\varepsilon \right)^{1/\varepsilon}.$$

By standard interpolation theory, L^ε norm is controlled by $L^{1,\infty}$ norm, i.e.

$$\left(\int_{B_{3R/2}} |\partial_{ee}^2 \tilde{w}_r|^\varepsilon \right)^{1/\varepsilon} \leq C(n, \varepsilon, R) \|\partial_{ee}^2 \tilde{w}_r\|_{L^{1,\infty}(B_{3R/2})}.$$

By Calderón–Zygmund theory, the right hand side is controlled by

$$C(n) (\|\Delta \tilde{w}_r\|_{L^1(B_{7R/4})} + \|\tilde{w}_r\|_{L^1(B_{7R/4})}).$$

Note that $\Delta \tilde{w}_r \leq 0$, pick any $\chi \in C_c^\infty(B_{2R})$ such that $0 \leq \chi \leq 1$ and $\chi = 1$ in $B_{7R/4}$, we obtain

$$\|\Delta \tilde{w}_r\|_{L^1(B_{7R/4})} \leq - \int_{B_{2R}} \chi \Delta \tilde{w}_r = - \int_{B_{2R}} \tilde{w}_r \Delta \chi \leq C(n, R) \|\tilde{w}_r\|_{L^1(B_{2R})}.$$

Finally, we have $\|\tilde{w}_r\|_{L^1(B_{2R})} \leq C(n, R) \|\tilde{w}_r\|_{L^2(B_2)} \leq C(n, R, \phi(1, u - p))$ by Lemma 3.5. Hence $\|(\partial_{ee}^2 \tilde{w}_r)_-\|_{B_R} \leq C$ as desired.

In addition, if $\dim\{p = 0\} = n - 1$, taking $\Delta \tilde{w}_r \leq 0$ into account, we see $\partial_{e'e'}^2 \tilde{w}_r \leq C$ for $e' \in S^{n-1} \cap \{p = 0\}^\perp$. Thanks to these semi-convexity and semi-concavity estimates, as well as locally boundedness of Corollary 3.7, we conclude² that $\|\nabla \tilde{w}_r\|_{L^\infty(B_{R/2})} \leq C(n, r, \phi(1, u - p))$. $\square$

Now we can provide the proof of Proposition 3.2. For any non-empty set $S \subseteq \mathbb{R}^n$, we denote with $B_r(S)$ the r -neighborhood of S , i.e. $B_r(S) = \cup_{s \in S} B_r(s)$.

²If this argument is not that obvious, consider a one dimensional case first. Suppose f is convex in $[-3, 3]$, bounded by C in $[-2, 2]$, then f' is bounded in $[-1, 1]$ by some constant depending on C .

Proof of Proposition 3.2. First, Lemma 3.5 together with a diagonal argument implies that $\tilde{w}_{r_k}$ converges weakly to some q in $H_{\text{loc}}^1(\mathbb{R}^n)$, up to a subsequence. What's more, the compactness of trace operator implies that $\|q\|_{L^2(\partial B_1)} = 1$ (and thus $q \neq 0$).

- (1) Suppose $0 \leq m \leq n-2$, we show that q is harmonic. After a change of coordinates $p_2 = a_1 x_1^2 + \dots + a_{n-m} x_{n-m}^2$, and $L = \{x \in \mathbb{R}^n \mid x_{n-m+1} = \dots = x_n = 0\}$. If $u(x) = 0$, since $u(x) = p_2(x) + o(|x|^2)$, we have $x_1, \dots, x_{n-m} = o(1)$ as $|x| \rightarrow 0$. That is, for any $\delta > 0$, as $r \rightarrow 0$, $\{u_r = 0\} \cap B_R \subseteq B_\delta(L) \cap B_R$ for any fixed $R > 0$. Since $\Delta w_r = -r^2 \chi_{\{u_r=0\}}$ is supported in $\{u_r = 0\}$, we obtain that q is harmonic in $\mathbb{R}^n \setminus L$. Finally, since L has codimension 2, q must be harmonic in $\mathbb{R}^n$, by a capacity argument³.

Next we show q is λ_2 -homogeneous, which is equivalent to $\phi(r, q) = 1$ for all $0 < r < 1$. The weak lower continuity of Dirichlet energy yields

$$\phi(1, q) \leq \liminf_{k \rightarrow \infty} \phi(1, \tilde{w}_{r_k}) = \liminf_{k \rightarrow \infty} (r_k, w) = \lambda_2.$$

To show $\phi(0+, q) \geq \lambda_2$, suppose on the contrary that $\phi(0+, q) < \lambda_2$, then $H(r, q) \gtrsim r^{\phi(0+, q) + \delta} \gg r^{\lambda_2}$ (this is the harmonic version of Lemma 2.5 and Remark 2.7). On the other hand, $H(r, \tilde{w}_{r_k}) \leq r^{2\lambda_2} H(1, \tilde{w}_{r_k}) = r^{2\lambda_2}$, hence $H(r, q) \leq r^{2\lambda_2}$. Contradiction! So $\lambda_2 \leq \phi(0+, q) \leq \phi(1, q) \leq \lambda_2$, and q is λ_2 -homogeneous.

Finally, homogeneous harmonic functions are just harmonic polynomials. Therefore $\lambda_2 \in \mathbb{N}$, and in fact $\lambda_2 \geq 2$ by Corollary 2.3. The rest follows by Lemma 3.4.

- (2) Now we turn to $m = n-1$, which is a little more involved. Lemma 3.8 implies $\tilde{w}_{r_k} \rightarrow r$ also in $C_{\text{loc}}^0(\mathbb{R}^n)$. As before, we have $\Delta q = 0$ in $\mathbb{R}^n \setminus L$, and $\Delta q \leq 0$ holds because $\Delta \tilde{w}_r \leq 0$. On $L = \{p_2 = 0\}$, we have $w_r = u_r \geq 0$. To conclude that q is a solution of the thin obstacle problem, it remains to check $q \Delta q = 0$.

Because $\phi(0+, w)$ exists, there exists $\bar{r}_k \in [r_k, 2r_k]$ such that $\bar{r}_k \phi'(\bar{r}_k, w) \rightarrow 0$. Recall $\phi'(r, w) \geq 2 \left(\int_{B_1} \tilde{w}_r \Delta \tilde{w}_r \right)^2$, we obtain that for $\rho_k := \bar{r}_k / r_k \in [1, 2]$,

$$\int_{B_1} \tilde{w}_{r_k} \Delta \tilde{w}_{r_k} \leq \int_{B_{\rho_1}} \tilde{w}_{r_k} \Delta \tilde{w}_{r_k} \leq C \int_{B_{\bar{r}_k}} \tilde{w}_{\bar{r}_k} \Delta \tilde{w}_{\bar{r}_k} \rightarrow 0.$$

Note that $\Delta w \leq 0$, thus $\Delta \tilde{w}_{r_k}$ is a Radon measure converging in the weak* sense to Δq . As $\tilde{w}_{r_k}$ converges to q uniformly, and $\tilde{w}_{r_k} \Delta \tilde{w}_{r_k} \geq 0$, one can conclude

$$\int_{B_1} q \Delta q = 0 \Rightarrow q \Delta q = 0.$$

Same arguments (where indices are more messy) would show $q \Delta q = 0$ in $\mathbb{R}^n$.

Next, since the frequency argument also holds for solutions of the thin obstacle problem (in fact, in deriving monotonicity formula, one only needs $q \Delta q = 0$ rather than $\Delta q = 0$), the same argument as the former cases would show q is λ_2 -homogeneous.

³Frankly speaking, I don't know what does this "capacity argument" mean. Here's a another approach (perhaps essentially the same): first prove directly that q is harmonic in distribution sense with the help of $\dim L \leq n-2$, then q is harmonic by Weyl's lemma. We point out this result fails if L has codimension 1. For example, x_+ is harmonic in $\mathbb{R} \setminus \{0\}$ but its Laplacian is δ_0 at 0.

Finally, to show $\lambda_2 \geq 2 + \alpha_0$ for some dimensional constant $\alpha_0 > 0$, we proceed by contradiction. Otherwise, there exists a sequence $\lambda_2^{(k)} \rightarrow 2$, and blow-ups $q^{(k)}$. Up to a subsequence, $q^{(k)}$ converges to $q^{(\infty)}$ in weak H^1 and strong C^0 , thus $q^{(\infty)}$ is 2-homogeneous. By Lemma 3.1, $q^{(\infty)}$ is a harmonic polynomial and $q^{(\infty)}$ is non-negative on L . Now (3) holds after a suitable change of coordinates. But this is impossible: $q^{(\infty)}$ being non-negative on L requires $N \leq 0$, contradictory to $\text{tr}(N) = t > 0$! $\square$

Here's an interesting corollary in dimension 2:

Corollary 3.9. *Let $n = 2$ and $0 \in \Sigma_1$, then*

$$\lambda_2 \in \{3, 4, 5, \dots\} \cup \{7/2, 11/2, 15/2, \dots\}$$

. In particular, $\lambda_2 \geq 3$.

Proof. Combining $\lambda_2 \geq 2 + \alpha_0$ and the possible homogeneity of the solution of thin obstacle problem in Lemma 3.1. $\square$

We end this subsection with an interesting (and useful) observation that $\lambda_2 - 2$ controls the measure of $\{u = 0\} \cap B_r$. We point out that for $m \leq n - 2$ and $m = n - 1$, this result is strengthened in different manners. We shall use frequently the case $m = n - 1$.

Lemma 3.10. *Let $0 \in \Sigma(u)$, $w = u - p_2$ and $\lambda_2 = \phi(0+, w)$, then*

$$\frac{|\{u = 0\} \cap B_r|}{|B_r|} \leq Cr^{\lambda_2 - 2},$$

where C depends on $n, \phi(1, u - p_2)$. Moreover, let $m := \dim\{p_2 = 0\}$, if $0 \leq m \leq n - 2$, then this C can be replaced by $o(1)$; if $m = n - 1$, then

$$\text{dist}(x, L) \leq Cr^{\lambda_2 - 1}, \quad \forall x \in \{u = 0\} \cap B_r.$$

In particular, since $\lambda_2 \geq 2 + \alpha_0$, we have $\{u = 0\} \cap B_r \subseteq B_{Cr^{1+\alpha_0}}(L) \cap B_r$.

Proof. By the same argument in Lemma 3.8, $\int_{B_{1/2}} |\Delta \tilde{w}_r| \leq C \|\tilde{w}_r\|_{L^1(B_1)} \leq C$. Since

$$|\Delta \tilde{w}_r| = r^2 \frac{\chi_{\{u_r=0\}}}{H(r, w)} \geq r^{2-\lambda_2} \chi_{\{u_r=0\}},$$

we conclude that $|\{u_r = 0\} \cap B_{1/2}| \leq Cr^{\lambda_2 - 2}$ as desired.

Moreover, if $0 \leq m \leq n - 2$, then $\Delta \tilde{w}_r \leq 0$ and $\tilde{w}_r$ converges to a harmonic function, it follows that $\int_{B_{1/2}} |\Delta \tilde{w}_r| \rightarrow 0$. And if $m = n - 1$, we can assume $p_2 = \frac{1}{2}x_n^2$, hence $\{p_2 = 0\} = \{x_n = 0\}$. We have $|\nabla \tilde{w}_r| \leq C$ in $B_{1/2}$, or

$$|\nabla u(x) - \nabla p_2(x)| \leq C \frac{H(r, w)}{r} \leq Cr^{\lambda_2 - 1}, \quad \forall x \in B_{r/2}.$$

Note that if $x \in \{u = 0\}$, then $\nabla u(x) = 0$ and $|\nabla p_2(x)| = |x_n| = \text{dist}(x, \{p_2 = 0\})$. $\square$

3.3 Proof of Theorem 1.1(2)

In this subsection we prove Theorem 1.1(2) involving only the lower stratum, which is much easier than Σ_{n-1} . We leave the proof of Theorem 1.1(1) after studying Σ_{n-1} in detail in the upcoming sections. The first lemma, however, doesn't require $\dim\{p_2 = 0\} \leq n - 2$.

Lemma 3.11. *Suppose $0 \in \Sigma(u)$, $r_l \rightarrow 0$ and $0 \neq x_l \in \Sigma(u) \cap \overline{B_{r_l}}$ satisfies $x_l \rightarrow 0$. If $\tilde{w}_l := (u - p_2)_r / H(r, u - p_2)^{1/2} \rightharpoonup q$ in $H_{loc}^1(\mathbb{R}^n)$ and $y_l := x_l / r_l \rightarrow y_\infty$, then $y_\infty \in \{p_2 = 0\}$ and $q(y_\infty) = 0$.*

Proof. Since $u - p_2 = o(|x|^2)$ and $u(r_l y_l) = u(x_l) = 0$, we have $r_l^2 p_2(y_l) = p_2(r_l y_l) = o(r_l^2)$ as $l \rightarrow \infty$, hence $p_2(y_l) = 0$.

Recall q is λ_2 -homogeneous with $\lambda_2 \geq 2$ (Proposition 3.2), to show $q(y_\infty) = 0$, we may assume $y_\infty \neq 0$. We know that $\phi(1/2, u(r_l(y_l + \cdot)) - p_2(r_l \cdot)) \geq 2$. Rewrite this as

$$2 \leq \frac{1}{2} \frac{\int_{B_{1/2}} |\nabla \tilde{w}_{r_l}(y_l + \cdot) + h_{r_l}^{-1} \nabla (p_2(r_l(y_l + \cdot)) - p_2(r_l \cdot))|^2}{\int_{\partial B_{1/2}} |\tilde{w}_{r_l}(y_l + \cdot) + h_{r_l}^{-1} (p_2(r_l(y_l + \cdot)) - p_2(r_l \cdot))|^2}, \quad (5)$$

where $h_{r_l} := H(r_l, u - p_2)$. Since p_2 is a quadratic polynomial, we have

$$h_{r_l}^{-1} (p_2(r_l(y_l + x)) - p_2(r_l x)) = c_l + b_l \cdot x,$$

for some $c_l \in \mathbb{R}$ and $b_l \in \mathbb{R}^n$ with $b_l \perp \{p_2 = 0\}$.

We claim that $|c_l|, |b_l| \leq C$ for some constant C independent of l . Otherwise, note that $\tilde{w}_{r_l}$ is locally bounded in H^1 , divide by $(|b_k| + |c_k|)^2$ in numerator and denominator of (5), we obtain

$$2 \leq \frac{1}{2} \frac{\int_{B_{1/2}} |o(1) + \bar{b}_l|^2}{\int_{\partial B_{1/2}} |o(1) + \bar{c}_l + \bar{b}_l \cdot x|^2},$$

where $\bar{c}_l := c_l / (|c_l| + |b_l|)$ and $\bar{b}_l := b_l / (|c_l| + |b_l|)$. Up to a subsequence, we have $\bar{c}_l \rightarrow \bar{c}_\infty$, $\bar{b}_l \rightarrow \bar{b}_\infty$, and

$$2 \leq \frac{1}{2} \frac{\int_{B_{1/2}} |\bar{b}_\infty|^2}{\int_{\partial B_{1/2}} |\bar{c}_\infty + \bar{b}_\infty \cdot x|^2} = \frac{1}{2} \frac{2^{-n} \omega_n |\bar{b}_\infty|^2}{2^{1-n} n \omega_n |\bar{c}_\infty|^2 + 2^{-1-n} \omega_n |\bar{b}_\infty|^2} = \frac{|\bar{b}_\infty|^2}{4n |\bar{c}_\infty|^2 + |\bar{b}_\infty|^2} \leq 1.$$

This contradiction proves the claim.

Up to a subsequence, we have $c_l \rightarrow c_\infty$ and $b_l \rightarrow b_\infty$ with $b_\infty \perp \{p_2 = 0\}$. By Remark 2.6, we have

$$\rho^{-4} \oint_{\partial B_\rho} |\tilde{w}_{r_l}(y_l + x) + c_k + b_l \cdot x|^2 \leq 2^4 \oint_{\partial B_{1/2}} |\tilde{w}_{r_l}(y_l + x) + c_l + b_l \cdot x|^2.$$

Taking limit, we obtain

$$\rho^{-4} \oint_{\partial B_\rho} |q(y_\infty + x) + c_\infty + b_\infty \cdot x|^2 \leq 2^4 \oint_{\partial B_{1/2}} |q(y_\infty + x) + c_\infty + b_\infty \cdot x|^2.$$

The left hand is bounded as $\rho \rightarrow 0$, so we must have $\nabla q(y_\infty) = -b \in \{p_2 = 0\}^\perp$. By homogeneity we get $q(y_\infty) = 0$. $\square$

Definition 3.12. For $m = 1, \dots, n-1$, define

$$\Sigma_m^a := \{x \in \Sigma_m \mid \lambda_2 = \phi(0+, u(x_0 + \cdot) - p_{2,x_0}) < 3\},$$

and $\Sigma_m^g := \Sigma_m \setminus \Sigma_m^a$.

Recall $x \mapsto \lambda_{2,x}$ is upper-semicontinuous⁴, Σ_m^a is relatively open in Σ_m .

Lemma 3.13. For $n \geq 3$, Σ_1^a is discrete in $\Sigma(u)$ (hence it's countable).

Proof. Suppose on the contrary that $0 \in \Sigma_1^a$ and $\Sigma(u) \ni x_l \rightarrow 0$. Let $r_l = |x_l|$ and $y_l = x_l/r_l$. By Proposition 3.2, up to a subsequence, we have $\tilde{w}_{r_l} \rightarrow q$ in $H_{\text{loc}}^1(\mathbb{R}^n)$ for some 2-homogeneous function q (for $n \geq 3$ and $0 \in \Sigma_1^a$ implies $\lambda_2 = 2$). Suppose, up to a sub-subsequence, $y_l \rightarrow y_\infty \in \partial B_1$, then by Lemma 3.11 we have $p(y_\infty) = q(y_\infty) = 0$, thus $\{p = 0\} = \mathbb{R}y_\infty$. Now (3) holds in a suitable coordinate system. But this is impossible because we have $N = 0$ and $\text{tr } N > 0!$ $\square$

Before proving a similar result for Σ_m^a ($m = 2, \dots, n-2$), we need a lemma from geometric measure theory.

Lemma 3.14. Let $E \subseteq \mathbb{R}^n$ be a set with $\mathcal{H}^\beta(E) > 0$ for some $\beta > 0$, then:

(1) For $\mathcal{H}^\beta$ -a.e. $x_0 \in E$, the “upper density”

$$\limsup_{r \rightarrow 0} \frac{\mathcal{H}_\infty^\beta(E \cap B_r(x_0))}{\omega_\beta r^\beta} \geq c_n > 0,$$

where $\omega_\beta = \pi^{\beta/2} \Gamma(\beta/2 + 1)$. Such points are called “density points”.

(2) Assume x_0 is a density point, let $r_l \rightarrow 0$ such that

$$\lim_{l \rightarrow \infty} \frac{\mathcal{H}_\infty^\beta(E \cap B_{r_l}(x_0))}{\omega_\beta r_l^\beta} \geq c_n.$$

For any $R > 0$, define the “accumulation set”

$$\mathcal{A}_{E, x_0, \{r_l\}, R} := \{x \in \overline{B_R} \mid \exists x_l \in r_l^{-1} E \cap B_R, \text{ such that } x_l \rightarrow x\}.$$

Then $\mathcal{H}^\beta(\mathcal{A}_{E, x_0, \{r_l\}, R}) > 0$.

Proof. (1) This is well-known, and we can take $c_n = 2^{-n}$. See e.g. [7, Theorem 3.26].

(2) Assume on the contrary that $\mathcal{H}^\beta(\mathcal{A}_{E, x_0, \{r_l\}, R}) = 0$, then for any $\varepsilon > 0$, there exists an open set U containing $\mathcal{A}_{E, x_0, \{r_l\}, R}$. It's easy to verify that $r_l^{-1} E \cap B_R \subseteq U$ for $l \gg 1$, thus

$$\lim_{l \rightarrow \infty} \frac{\mathcal{H}_\infty^\beta(E \cap B_{r_l}(x_0))}{\omega_\beta r_l^\beta} \leq \frac{\varepsilon}{\omega_\beta}.$$

Contradictory to the arbitrariness of ε and the fact that this limit $\geq c_n$. $\square$

⁴This is a standard argument: $\phi(r, u - p_{2,x})$ is continuous in x , and decreases to $\lambda_{2,x}$, hence upper-semicontinuous.

Lemma 3.15. *For $n \geq 4$ and $m = 2, \dots, n-2$, we have $\dim_{\mathcal{H}}(\Sigma_m^a) \leq m-1$.*

Proof. Otherwise, for some $\beta > m-1$, we have $\mathcal{H}_{\infty}^{\beta}(\Sigma_m^a) > 0$. Then there's a density point $x_0 \in \Sigma_m^a$. Let $r_l \rightarrow 0$ be the associated radius such that $r_l^{-\beta} \mathcal{H}_{\infty}^{\beta}(\Sigma_m^a \cap B_{r_l}(x_0)) > c_{n,\beta}$. Up to a subsequence, we have $\tilde{w}_{r_l} \rightarrow q$ in $H_{\text{loc}}^1(\mathbb{R}^n)$ with q a 2-homogeneous harmonic polynomial. We can assume in appropriate coordinate system (3) holds for p_2 and q .

Consider the accumulation set $\mathcal{A} := \mathcal{A}_{\Sigma_m^a, x_0, \{r_l\}, 1}$. If $x \in \mathcal{A}$, then there exists a subsequence $x_{l_j} \in \Sigma_m^a$ such that $x_{l_j} \leq r_{l_j}$ and $x_{l_j}/r_{l_j} \rightarrow x$. By Lemma 3.11, we find that $\mathcal{A} \subseteq \overline{B_1} \cap \{p_2 = 0\} \cap \{q = 0\}$. However, $\dim(\{p_2 = 0\} \cap \{q = 0\}) \leq m-1$ by (3), so $\mathcal{H}^{\beta}(\mathcal{A}) = 0$, contradictory to Lemma 3.14(2). $\square$

In order to prove Theorem 1.1, we need the following version of Whitney's extension theorem. For a proof, see e.g. [4, Appendix C].

Lemma 3.16. *Let $\beta \in (0, 1]$, $l \in \mathbb{N}$, $K \subseteq \mathbb{R}^n$ be a compact set, and $f: K \rightarrow \mathbb{R}$. Suppose for each $x_0 \in K$, there exists a polynomial P_{x_0} of degree l , such that $p_{x_0}(x_0) = f(x_0)$ and*

$$|\nabla^k P_{x_0}(x) - \nabla^k P_x(x)| \leq C|x - x_0|^{l+\beta-k}, \quad \forall x \in K, \quad k \in \{0, 1, \dots, l\},$$

where C is independent of x_0 . Then there exists a $C^{l,\beta}$ function $F: \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$F|_K = f \quad \text{and} \quad F(x) = P_{x_0}(x) + O(|x - x_0|^{l+\beta}).$$

Remark 3.17. *Since all norms on the space of all polynomials of degree $\leq l$ are equivalent, it's sufficient to check*

$$\|(P_{x_0} - P_x)(\rho \cdot)\|_{L^2(B_1(x_0))} \leq C\rho^{\lambda},$$

where $\rho = |x_0 - x|$ and C is uniform for all $x_0, x \in K$.

Remark 3.18. *In fact, this extension theorem can be modified for other modulus of continuity. The version above is for $\omega(r) = r^{\beta}$. Another version, in which $\omega(r) = \log^{-\varepsilon_0}(r)$, says if*

$$\|(P_{x_0} - P_x)(\rho \cdot)\|_{L^2(B_1(x_0))} \leq C\rho^l \log^{-\varepsilon_0}(|x - x_0|),$$

then $F \in C^{l, \log^{-\varepsilon_0}}(\mathbb{R}^n)$.

Applying this extension theorem, we easily derive the following lemma, from which the main part of Theorem 1.1(2) follows easily.

Lemma 3.19. *Let $m \in \{1, 2, \dots, n-1\}$ and $\lambda > 2$. If $\beta \in (0, 1]$ and $l \in \mathbb{N}$ satisfies $\lambda = l + \beta$, then⁵*

$$S_{m,\lambda} := \{x_0 \in \Sigma_m \mid \phi(0+, u(x_0 + \cdot) - P_{2,x_0}) \geq \lambda\}$$

is locally contained in a m -dimensional manifold of class $C^{l,\beta}$.

Proof. We can assume $0 \in \Sigma_{m,\lambda}$ and prove this result near 0. Define $S_{\lambda} := \{x_0 \in \Sigma(u) \mid \phi(0+, u(x_0 + \cdot) - p_{2,x_0}) \geq \lambda\}$, then S_{λ} is closed since $\Sigma(u)$ is closed and $x \mapsto \phi(0+, u(x + \cdot) - p_{2,x})$ is upper-semicontinuous. Let the compact set $K := S_{\lambda} \cap \overline{B_{1/8}}$, then $\overline{S_{m,\lambda}} \cap B_{1/8} \subseteq K$.

⁵The notations $S_{m,\lambda}$ and S_{λ} are not the same as the set of all homogeneous solutions of the thin obstacle problem, and only appear in this lemma and its proof.

For each $x_0 \in K$, let $P_{x_0}(x) = p_{2,x_0}(x - x_0)$. We shall check that the assumptions of Lemma 3.16 are satisfied for $K, f \equiv 0$ and P_{x_0} . Clearly, $P_{x_0}(x_0) = 0 = f(x_0)$ is satisfied. For any $x_0, x \in K$, let $\rho = |x - x_0|$, then (note that $\rho \leq 1/4$, so that the functions below are defined in $B_2(x)$)

$$\begin{aligned} \|(P_{x_0} - P_x)(\rho)\|_{L^2(B_1(x_0))} &\leq \|u(\rho) - P_{x_0}(\rho)\|_{L^2(B_1(x_0))} + \|u(x_0) - P_x(\rho)\|_{L^2(B_1(x))} \\ &= \|u(\rho) - p_{2,x_0}(\rho \cdot -x_0)\|_{L^2(B_1(x_0))} + \|u(\rho) - p_{2,x}(\rho \cdot -x)\|_{L^2(B_1(x_0))} \\ &\leq \|u(x_0 + \rho) - p_{2,x_0}(\rho)\|_{L^2(B_1)} + \|u(\rho) - p_{2,x}(\rho \cdot -x)\|_{L^2(B_2(x))} \\ &= \|u(x_0 + \rho) - p_{2,x_0}(\rho)\|_{L^2(B_1)} + \|u(x + \rho) - p_{2,x}(\rho)\|_{L^2(B_2(x))}. \end{aligned}$$

By Remark 2.7, we have $H(r, u(x_0 + \cdot) - p_{2,x_0}) \leq C(n, \phi(1/4), u(x_0 + \cdot) - p_{2,x_0})r^\lambda$, hence

$$\|u(x_0 + \rho) - p_{2,x_0}(\rho)\|_{L^2(B_1)} \leq C\rho^\lambda,$$

where C is independent of x_0 . The same estimate holds for $\|u(x + \rho) - p_{2,x}(\rho)\|_{L^2(B_2(x))}$, so

$$\|(P_{x_0} - P_x)(\rho)\|_{L^2(B_1(x_0))} \leq C\rho^\lambda.$$

We are done by Remark 3.17.

By Whitney's extension theorem, we find a function $F \in C^{l,\beta}(\mathbb{R}^n)$, such that $F(x) = p_{2,x_0}(x - x_0) + O(|x - x_0|^{l+\beta})$ for all $x_0 \in K$. Therefore

$$S_{m,\lambda} \cap B_{1/4} \subseteq K \subseteq \{\nabla F = 0\}.$$

Finally, since the rank of $\nabla^2 F = \nabla^2 p_{2,x_0}$ is m for $x_0 \in \Sigma_m$, we are done. $\square$

Proof of Theorem 1.1(2). Except for Σ_m is locally contained in a $C^{1,\log^{\varepsilon_0}}$ manifold, other statements are contained in Lemma 3.13, Lemma 3.15 and Lemma 3.19. By Remark 3.18, we only need to show

$$\|u(x_0 + r\cdot) - p_{2,x_0}(r\cdot)\|_{L^2(\partial B_1)} \leq Cr^2 \log^{-\varepsilon_0}(1/r), \quad \forall x_0 \in \Sigma_m \cap B_{1/2}, \quad r \in (0, 1/2). \quad (6)$$

The rest follows exactly as Lemma 3.19.

Recall Caffarelli [1] showed that, for some dimensional constant ε_0, C_0 and any direction $e \in S^{n-1}$, we have

$$\partial_{ee}u(x) \geq -C_0 \log^{-\varepsilon_0}(1/r), \quad \forall x \in B_r(x_0), \quad x_0 \in \Sigma(u).$$

If (6) fails for $C = M$ (to be determined later), then there exists a sequence $r_l \rightarrow 0$ such that $\partial_{ee}\tilde{w}_{r_l} \geq -C_0/M$. Passing to limit we obtain $\partial_{ee}q \geq -C_0/M$. However, by $\text{tr}(\nabla^2 q)$ and $\|q\|_{L^2(\partial B_1)} = 1$, there exists a $c_1(n)$ such that $\inf_{e \in S^{n-1}} \partial_{ee}q \leq -c_1$. Contradictory if we choose $M = C_0/c_1$. $\square$

4 Lipschitz estimates for $u - P_k$

From now on, we will focus on Σ_{n-1} , for our targets are Theorem 1.1(1) and Theorem 1.4, both concerning only Σ_{n-1} (recall $\dim \Sigma_m \leq m \leq n-2$ for $m \leq n-2$). So we assume $0 \in \Sigma_{n-1}$.

In this section, we make some preparations for studying higher order blow-ups. We've examined the second order blow-up $\tilde{w}_r = (u - p_2)_r / H(r, u - p_2)^{1/2}$ in Proposition 3.2. Remark 2.7 yields that, if $\lambda_2 > 3$, then $H(r, u - p_2) \ll r^3$. Therefore, as $r \rightarrow 0$,

$$r^{-3}(u - p_2)_r \rightarrow 0, \quad \text{provided } \lambda_2 > 3.$$

We can view this as the third-order blow-up⁶. Now the question arises that what would happen if $\lambda_2 \leq 3$. Also due to Remark 2.7, we can't expect this limit exists in general if $\lambda_2 < 3$. Then the main question is the case $\lambda_2 = 3$. This third order blow-up was first analysed in detail in [3, Section 4]. They proved that, for most points in Σ_{n-1} , the third order blow-up is a 3-homogeneous harmonic polynomial p_3 that vanishes on $\{p_2 = 0\}$. The key ingredient was the “almost monotonicity” of $H_3(r, u - p_2 - q)$ for any 3-homogeneous harmonic polynomial vanishing on $\{p_2 = 0\}$.

After that, [2] examined the fourth order blow-up. Intuitively, suppose the third order blow-up p_3 exists and is a 3-homogeneous harmonic polynomial, then one should look at $r^{-4}(u - p_2 - p_3)$. However, this time even the almost monotonicity fails. To overcome this difficulty, they observed that one should consider $r^{-4}(u - P_3)$ instead, where P_3 is another polynomial determined by p_2, p_3 . Then they proved an Lipschitz estimate by considering derivatives with respect to a rotational vector field, taking advantage that Δ commutes with rotations, and finally recovered an almost monotonicity formula. With this in hand, they can continue to study blow-ups to fifth order.

But their approach ended at the fifth order. It was [4] that took a different approach to prove the Lipschitz estimates, and examined the blow-ups to arbitrary order. They figured out how to construct P_k with blow-up limits $p_2, \dots, p_k$ in hand. The main targets of this section is to construct P_k and prove Lipschitz estimates for $u - P_k$. We'll leave the almost monotonicity formulae and the study of higher order blow-ups to next section.

4.1 Polynomial Ansätze

Denote with V_k the vector space consisting of all k -homogeneous polynomials (and 0). Projection $\pi_k: \mathbb{R}[x] \rightarrow V_k$ sends a polynomial to its k -homogeneous part, and $\pi_{\leq k} = \pi_0 + \pi_1 + \dots + \pi_k$.

Definition 4.1. For $k \geq 2$, $P_k \subseteq V_2 \times \dots \times V_k$ consists of $(p_2, \dots, p_k)$ with the following properties:

- (1) $p_2 \geq 0$, $\Delta p_2 = 1$, $\dim\{p_2 = 0\} = n - 1$ (thus $p_2 = \frac{1}{2}x_n^2$ after an appropriate rotation);
- (2) $\Delta p_j = 0$ and p_j vanishes on $\{p_2 = 0\}$ for $3 \leq j \leq k$.

Note that $p_2 = \frac{1}{2}(v \cdot x)^2$ for some $v \in S^{n-1}$, up to changing v to $-v$. We shall equip $P_k \subseteq V_2 \times \dots \times V_k$ with a norm by

$$|(p_2, \dots, p_k)| := \sum_{j=2}^k \|p_j\|_{L^2(\partial B_1)}.$$

Note that since this is a finite dimension space, all norms are equivalent.

Lemma 4.2. For $p_2 \in P_2$, the map $\delta: V_k \rightarrow V_k$ defined by $\delta(q) = \Delta(p_2 q)$ is an isomorphism.

Proof. We only need to prove δ is injective. Since Δ commutes with rotations, we can assume $p_2(x) = \frac{1}{2}x_n^2$. In fact, we can prove that $\Delta(x_n^m q) \neq 0$ if $m \geq 2$ and $x_n \nmid q$. If not, we obtain

$$m(m-1)x_n^{m-2}q + 2mx_n^{m-1}\partial_n q + x_n^m \Delta q = 0.$$

And this implies $x_n \mid q$. □

⁶If u has a Taylor expansion $u(x) = p_2(x) + p_3(x) + o(|x|^3)$ for some 3-homogeneous polynomial p_3 , then $r^{-3}(u - p_2)_r \rightarrow p_3$. Since our goal is to prove most points in Σ_{n-1} have expansions to arbitrary order, studying such type of blow-up is natural and reasonable.

Lemma 4.3. For $k \geq 2$ and any $(p_2, \dots, p_k) \in P_k$, there exists a unique collection $(R_1, \dots, R_{k-1}) \subseteq V_1 \times \dots \times V_{k-1}$, such that: if $p_2 = \frac{1}{2}(v \cdot x)^2$ with $|v| = 1$, then

$$A_{k,v}(x) := v \cdot x + \sum_{j=1}^{k-1} (v \cdot x) R_j(x) + \sum_{j=3}^k \frac{p_j(x)}{v \cdot x}$$

satisfies $\Delta \left(\frac{1}{2} A_{k,v}^2 \right) = 1 + O(|x|^k)$. In fact, we always have $R_1 = 0$ and $A_{2,v} = v \cdot x$. Furthermore, R_j is determined by $p_2, \dots, p_{j+1}$, and doesn't depend on choosing v or $-v$, so that $A_{k,-v} = -A_{k,v}$.

Proof. We make an induction on k . When $k = 2$, it says $A_{2,v} = v \cdot x + (v \cdot x) R_1(x)$, hence

$$\Delta \left(\frac{1}{2} A_{2,v}^2 \right) = 1 + \Delta(2p_2 R_1) + O(|x|^2).$$

Since $\Delta(p_2 R_1)$ has degree 1 unless it's identically 0, we must have $\Delta(p_2 R_1) = 0$. By Lemma 4.2, this is equivalent to $R_1 = 0$, and thus $A_{2,v} = v \cdot x$. Suppose the statement holds for $k \geq 2$. Since $A_{k+1,v} = A_{k,v} + (v \cdot x) R_k + \frac{p_{k+1}}{v \cdot x}$, we have (recall $\Delta p_{k+1} = 0$)

$$\Delta \left(\frac{1}{2} A_{k+1,v}^2 \right) = \Delta \left(\frac{1}{2} A_{k,v}^2 \right) + \Delta(2p_2 R_k) + \Delta(p_{k+1} R_1) + \Delta \left(\frac{p_3 p_{k+1}}{2p_2} \right) + O(|x|^{k+1}).$$

So $\Delta \left(\frac{1}{2} A_{k+1,v}^2 \right) = 1 + O(|x|^{k+1})$ if and only if

$$\Delta \left(\frac{1}{2} A_{k,v}^2 \right) = 1 + O(|x|^k) \quad \text{and} \quad \pi_k \left(\Delta \left(\frac{1}{2} A_{k,v}^2 \right) \right) + \Delta(2p_2 R_k) + \Delta(p_{k+1} R_1) + \Delta \left(\frac{p_3 p_{k+1}}{2p_2} \right) = 0.$$

Our inductive assumption gives $R_1 = 0$ and $R_2, \dots, R_k$ are uniquely determined. Again Lemma 4.2 implies there's a unique R_k , namely

$$R_k = -\frac{1}{2} \delta^{-1} \left(\pi_k \left(\Delta \left(\frac{1}{2} A_{k,v}^2 \right) \right) + \Delta \left(\frac{p_3 p_{k+1}}{2p_2} \right) \right). \quad \square$$

Definition 4.4. Given $k \geq 2$, we define $P_k: P_k \rightarrow \mathbb{R}[x]$ by

$$P_k(p_2, \dots, p_k) = \pi_{\leq k+1} \left(\frac{1}{2} A_{k,v}^2 \right),$$

where $p_2 = \frac{1}{2}(v \cdot x)^2$.

Recall $A_{k,-v} = -A_{k,v}$, hence P_k doesn't depend on the choice of v . Sometimes for simplicity we drop v and simply write A_k . We gather together the main properties of P_k .

Lemma 4.5. Let $k \geq 2$, $(p_2 = \frac{1}{2}(v \cdot x)^2, \dots, p_k) \in P_k$, and $\tau > 0$ such that $|(p_2, \dots, p_k)| \leq \tau$.

- (1) $\Delta P_k = 1$.
- (2) For any $e \in S^1$, $\partial_e P_k = \partial_e \left(\frac{1}{2} A_k^2 \right) + O(|x|^{k+1})$.
- (3) When $k \geq 3$, $P_k(p_2, \dots, p_k) = P_{k-1}(p_2, \dots, p_{k-1}) + p_k + O(|x|^{k+1})$.

(4) For some $r_0 = r_0(n, k, \tau) \in (0, 1)$, when $|x| \leq r_0$, we have $\frac{1}{2} \leq |\partial_v A_{k,v}(x)| \leq 2$, thus

$$\frac{1}{2}|A_k(x)| \leq \left| \partial_v \left(\frac{1}{2} A_k^2(x) \right) \right| \leq 2|A_k(x)|.$$

(5) If $r^{-2}u_r \rightarrow p_2$, then for all $r \in (0, 1/2)$, we have

$$\sup_{B_r \cap \{u=0\}} |\partial_v P_k| \leq C r^{1+\alpha_0},$$

where α_0 is as in Proposition 3.2, and $C = C(n, k, \tau)$.

Proof. (1) $\Delta P_k = \Delta(\frac{1}{2}A_k^2) + O(|x|^k) = 1 + O(|x|^k)$, and $\deg \Delta P_k \leq k-1$.

(2) It follows directly from $P_k = \frac{1}{2}A_k^2 + O(|x|^{k+2})$.

(3) Since $A_{k,v}(p_2, \dots, p_k) = A_{k-1,v} + (v \cdot x)R_{k-1} + \frac{p_k}{v \cdot x}$, we find that for $k \geq 3$,

$$P_k = \frac{1}{2}A_{k-1,v}^2 + \frac{p_k^2}{4p_2} + (v \cdot x)R_{k-1}A_{k-1,v} + \frac{p_k A_{k-1,v}}{v \cdot x} + O(|x|^{k+2}) = P_{k-1} + p_k + O(|x|^{k+1}).$$

(4) It follows from $\partial_v A_{k,v} = 1 + O(|x|)$.

(5) Note that $P_k = p_2 + O(|x|^3)$ and $|\partial_v p_2(x)| = \text{dist}(x, \{p_2 = 0\})$, then apply Lemma 3.10. $\square$

4.2 Lipschitz estimates

We fix our setting for this subsection. $k \geq 2$ is an integer, $\tau > 0$, $\beta \in (0, \frac{\alpha_0}{k+2})$ (recall α_0 comes from Proposition 3.2), $(p_2, \dots, p_k) \in P_k$, $|(p_2, \dots, p_k)| \leq \tau$, and $r^2 u(r \cdot) \rightarrow p_2 = \frac{1}{2}x_n^2$. Set $v = u - P_k(p_2, \dots, p_k)$. Note that we only require p_2 is the blow-up of u , and no extra restriction to $p_3, \dots, p_k$ in this section. We will choose later that $\beta = \frac{\alpha_0}{2(k+2)}$, so it is determined by n .

Proposition 4.6. *The setting is as above.*

(1) There exists C, r_0, θ depending on n, k, τ , such that for $j = 1, \dots, n-1$ and $r \in (0, r_0)$, we have

$$\|\partial_j v_r\|_{L^\infty(B_1)} \leq C(\|v_{\theta r}\|_{L^2(B_2 \setminus B_{1/2})} + r^{k+2}).$$

(2) For $\beta \in (0, \frac{\alpha_0^2}{k+2})$, there exists C, r_0, θ depending on n, k, τ, β , such that for $r \in (0, r_0)$, we have

$$\|\partial_n v_r\|_{L^\infty(B_1)} \leq C(\|v_{\theta r}\|_{L^2(B_2 \setminus B_{1/2})} + r^{k+2})^{1-\beta}.$$

We remark that [2] proved for $\beta = 0$ when $k = 3$. In general, we need $\beta \in (0, \frac{\alpha_0^2}{k+2})$. Later we'll set $\beta = \frac{\alpha_0^2}{2(k+2)}$, so that the constants are determined by n, k, τ . The proof is divided into several steps. We show that L^∞ norm can be controlled by L^2 norm, which is similar to Lemma 3.6. Then the key is we can link ∇v_r to $\partial_n P_k$ restricted to $\{u = 0\}$ and provide bilateral control of their L^∞ norms. Finally some iteration argument applies, and we can verify Proposition 4.6.

Lemma 4.7. *In the setting above, for some $C = C(n, k, \tau)$, we have*

$$\|v_r\|_{L^\infty(B_1)} \leq C\|v_r\|_{L^2(B_2 \setminus B_{1/2})} + Cr^{k+2}, \quad \forall r \in (0, 1/2).$$

Proof. Since $\Delta v = -\chi_{\{u=0\}} \leq 0$, by mean-value inequality, for some $z \in \partial B_r$, we have

$$\min_{B_r} v = v(z) \geq \int_{B_{1/2}(z)} v_r \geq -C\|v_r\|_{L^2(B_2 \setminus B_{1/2})}.$$

To get the bound from above, note that inside $\{u = 0\} \cap B_r$ we have

$$v = -P_k = -\frac{1}{2}A_k^2 + O(|x|^{k+2}) \leq Cr^{k+2}, \quad C = C(n, k, \tau).$$

Hence $\max\{v, Cr^{k+2}\}$ is subharmonic and we get the upper bound with Cr^{k+2} added. $\square$

Lemma 4.8. *In the setting above, there exists $C = C(n, k, \tau)$, such that for $j = 1, \dots, n-1$ and $0 < r < 1/2$, we have*

$$\begin{aligned} \|\partial_j v_r\|_{L^\infty(B_1)} &\leq Cr^2 \sup_{\{u=0\} \cap B_r} |\partial_n P_k| + C\|v_r\|_{L^2(B_2 \setminus B_{1/2})} + Cr^{k+2}, \\ \|\partial_n v_r\|_{L^\infty(B_1)} &\leq Cr \sup_{\{u=0\} \cap B_r} |\partial_n P_k| + C\|v_r\|_{L^2(B_2 \setminus B_{1/2})} + Cr^{k+2}. \end{aligned}$$

Proof. Clearly we can assume $r < r_0$ for some r_0 depending on n, k, τ . For $j = 1, \dots, n-1$, $\partial_j v_r$ is harmonic in $B_r \cap \{u > 0\}$. In $B_r \cap \{u = 0\}$, we have $\partial_j v_r = -\partial_j P_k = -A_k \partial_j A_k + O(|x|^{k+1})$. Note that $A_k = x_n + O(|x|^2)$ and $\partial_j A_k \leq C|x|$, so we obtain

$$|\partial_j v_r| \leq C|x||A_k| + C|x|^{k+1}.$$

Also $\partial_n P_k = A_k \partial_n A_k = (1 + O(|x|))A_k$ for $|x| \ll 1$, hence

$$\sup_{B_r \cap \{u=0\}} |\partial_j v| \leq Cr \sup_{B_r \cap \{u=0\}} |\partial_n P_k| + Cr^{k+1} =: K.$$

Now $\max\{|\partial_j v|, K\}$ is subharmonic in B_r , hence for some $x \in \partial B_r$,

$$\sup_{B_r} |\partial_j v| \leq K + \frac{1}{r} \int_{B_{1/4}(x)} |\partial_j v_r| \leq K + \frac{1}{r} \|\partial_j v_r\|_{L^1(B_{5/4} \setminus B_{3/4})}.$$

Note that $\|\nabla v_j\|_{L^1(B_{5/4} \setminus B_{3/4})} \leq C(\|\Delta v_r\|_{L^1(B_{11/8} \setminus B_{5/8})} + \|v_r\|_{L^2(B_{11/8} \setminus B_{5/8})})$, and $\Delta v_r \leq 0$, take a non-negative cut-off function χ with $\chi = 1$ on $B_{11/8} \setminus B_{5/8}$ and supported in $B_2 \setminus B_{1/2}$, we get

$$\|\Delta v_r\|_{L^1(B_{11/8} \setminus B_{5/8})} \leq - \int \Delta v_r \chi = - \int v_r \Delta \chi \leq C\|v_r\|_{L^1(B_2 \setminus B_{1/2})} \leq C\|v_r\|_{L^2(B_2 \setminus B_{1/2})}.$$

Henceforth

$$\|\partial_j v_r\|_{L^\infty(B_1)} = r \sup_{B_r} |\partial_j v| \leq rK + C\|v_r\|_{L^2(B_2 \setminus B_{1/2})}.$$

The estimate for $|\partial_n v_r|$ follows by the same manner, except that we only have

$$|\partial_n v| \leq C|A_k| + C|x|^{k+1} \leq C|\partial_n P_k| + C|x|^{k+1}$$

without the extra $|x|$ in the front of $|\partial_n P_k|$. $\square$

Lemma 4.9. *In the setting above, for any $\beta \in (0, \frac{\alpha_0^2}{k+2})$, there exists $C = C(n, k, \tau, \beta)$, such that for all $0 < r < 1/2$, we have*

$$\sup_{B_{r/4} \cap \{u=0\}} |r \partial_n P_k| \leq C r^{\frac{\beta}{\alpha_0 - \beta}} \|\partial_n v_r\|_{L^\infty(B_1)} + C r^{\beta + \frac{\beta}{\alpha_0}} \|\nabla_{x'} v_r\|_{L^\infty}^{1 - \frac{\beta}{\alpha_0}} + C \|v_r\|_{L^\infty(B_1)} + C r^{k+2}.$$

Proof. (To be completed.) □

Proof of Proposition 4.6. (To be completed.) □

5 Higher order blow-ups

5.1 Monotonicity of truncated frequency

In this subsection, we show that $\phi^\gamma(u - P_k)$ is almost monotone for $\gamma < k+2$. Here p_2 is the blow-up limit of u , but $p_3, \dots, p_k$ can be arbitrary. We first prove the following lemma, then iterate.

Lemma 5.1. *Let $k \geq 2$, $\tau > 0$, $r^{-2}u_r \rightarrow p_2$, $(p_2, \dots, p_k) \in P_k$ and $|(p_2, \dots, p_k)| \leq \tau$, $v = u - P_k(p_2, \dots, p_k)$. For each $\gamma \in [0, k+2)$ and $\beta \in (0, \frac{\alpha_0^2}{k+2})$, set*

$$\varepsilon := \min\{\alpha_0 - (k+2)\beta, k+2 - \gamma\} > 0.$$

Then there exists C, r_0, θ depending on n, k, τ, β , such that for all $r \in (0, r_0)$, we have

$$\frac{d}{dr} \phi^\gamma(r, v) \geq -C r^{\varepsilon-1} (g^\gamma(r, v) + 1) (\phi^\gamma(r, v) + 1),$$

and

$$\frac{\int_{B_r} |v_r \Delta v_r|}{H(r, v) + r^{2\gamma}} \leq C r^\varepsilon (g^\gamma(r, v) + 1),$$

where

$$g^\gamma(r, v) := \frac{\|v_{\theta r}\|_{L^2(B_2 \setminus B_{1/2})}^2}{H(r, v) + r^{2\gamma}}.$$

Proof. Recall Lemma 2.8 on the derivative of truncated frequency function:

$$\frac{d}{dr} \phi^\gamma(r, v) \geq \frac{2}{r} \frac{\phi^\gamma(r, v) \int_{B_1} v_r \Delta v_r - \int_{B_1} (x \cdot \nabla v_r) \Delta v_r}{H(r, v) + r^{2\gamma}}.$$

Since Δv_r is supported in $\{u_r = 0\}$, we only need to control

$$\frac{2}{r} \frac{\|\phi^\gamma(r, v) v_r - x \cdot \nabla v_r\|_{L^\infty(B_1 \cap \{u_r=0\})} \int_{B_1 \cap \{u_r=0\}} |\Delta v_r|}{H(r, v) + r^{2\gamma}}.$$

For $x \in B_1 \cap \{u_r = 0\}$, we have the following estimates:

- $|x_n| \leq C r^{\alpha_0}$ (Lemma 3.10);

- $|v_r(x)| \leq Cr^{1+\alpha_0}|\partial_n P_k(rx)| + Cr^{k+2}$, in fact,

$$\begin{aligned}
|v_r(x)| &= \left| \frac{1}{2} A_k^2(rx) \right| + Cr^{k+2} & (P_k &= \frac{1}{2} A_k^2 + O(|x|^{k+2})) \\
&\leq Cr^{1+\alpha_0}|A_k(rx)| + Cr^{k+2} & (A_k &= x_n + O(|x|^2), |x_n| \leq Cr^{\alpha_0}) \\
&\leq Cr^{1+\alpha_0}|A_k(rx)||\partial_n A_k(rx)| + Cr^{k+2} & (\partial_n A_k &= 1 + O(|x|)) \\
&\leq Cr^{1+\alpha_0}|\partial_n P_k(rx)| + Cr^{k+2} & (\partial_n P_k &= A_k \partial_n A_k + O(|x|^{k+1}));
\end{aligned}$$

- for $j \neq n$, $\partial_j v_r(x) \leq Cr^2|\partial_n P_k(rx)| + Cr^{k+2}$ (see the proof of Lemma 4.8);
- $r|\partial_n P_k(rx)| \leq C(\|v_{\theta r}\|_{L^2(B_2 \setminus B_{1/2})} + r^{k+2})^{1-\beta}$ (Proposition 4.6);
- $\partial_n v_r(x) = r\partial_n P_k(rx)$ (by definition).

Putting them together, we obtain for all $x \in B_1 \cap \{u_r = 0\}$,

$$\begin{aligned}
|\phi^\gamma(v_r) - x \cdot \nabla v_r| &\leq (\phi^\gamma(r, v) + 1)(|v_r| + |\nabla_{x'} v_r| + r|x_n||\partial_n P_k(rx)|) \\
&\leq C(\phi^\gamma(r, v) + 1)(r^{1+\alpha_0}|\partial_n P_k(rx)| + r^{k+2}) \\
&\leq C(\phi^\gamma(r, v) + 1)r^{\alpha_0}(\|v_{\theta r}\|_{L^2(B_2 \setminus B_{1/2})} + r^{k+2})^{1-\beta} + C(\phi^\gamma(r, v) + 1)r^{k+2} \\
&\leq C(\phi^\gamma(r, v) + 1)r^\varepsilon(\|v_{\theta r}\|_{L^2(B_2 \setminus B_{1/2})} + r^{k+2}) + C(\phi^\gamma(r, v) + 1)r^{k+2},
\end{aligned}$$

where in the last line we simply used

$$(\|v_{\theta r}\|_{L^2(B_2 \setminus B_{1/2})} + r^{k+2})^\beta \geq r^{(k+2)\beta} \quad \text{and} \quad r^{\alpha_0 - (k+2)\beta} \leq r^\varepsilon.$$

Using $H(r, v) + r^{2\gamma} \geq r^{2\gamma}$ and $r^{k+2-\gamma} \leq r^\varepsilon$, we obtain

$$\frac{\|\phi^\gamma(r, v)v_r - x \cdot \nabla v_r\|_{L^\infty(B_1 \cap \{u_r=0\})}}{(H(r, v) + r^{2\gamma})^{1/2}} \leq Cr^\varepsilon(\phi^\gamma(r, v) + 1)((g^\gamma(r, v))^{1/2} + 1).$$

In the same manner, one easily get (in fact, it's contained in the inequalities above)

$$\frac{\|v_r\|_{L^\infty(B_1 \cap \{u_r=0\})}}{(H(r, v) + r^{2\gamma})^{1/2}} \leq Cr^\varepsilon((g^\gamma(r, v))^{1/2} + 1).$$

Therefore, to finish our proof, we only need to show

$$\frac{\int_{B_1 \cap \{u_r=0\}} |\Delta v_r|}{(H(r, v) + r^{2\gamma})^{1/2}} \leq C((g^\gamma(r, v))^{1/2} + 1).$$

This is easy. Recall $\Delta v_r = -r^2 \chi_{\{u_r=0\}} \leq 0$, so we can do as in Lemma 3.8 to obtain

$$\int_{B_1 \cap \{u_r=0\}} |\Delta v_r| \leq C(n) \|v_r\|_{L^2(B_2 \setminus B_1)},$$

then as in the proof of Lemma 4.6 to show right hand side is controlled by $C(\|v_{\theta r}\|_{L^2(B_2 \setminus B_{1/2})} + r^{k+2})$. The first term give rise to $(g^\gamma(r, v))^{1/2}$, and the second term $\leq r^{k+2-\gamma} \leq 1$. $\square$

Proposition 5.2. Let $k \geq 2$, $\tau > 0$, $r^{-2}u_r \rightarrow p_2$, $(p_2, \dots, p_k) \in P_k$ and $|(p_2, \dots, p_k)| \leq \tau$, $v = u - P_k(p_2, \dots, p_k)$. For each $\gamma \in [0, k+2)$, set $\beta = \frac{\alpha_0^2}{2(k+2)}$ and

$$\varepsilon := \min\{\alpha_0 - (k+2)\beta, k+2-\gamma\} > 0.$$

Then there exists C, r_0, θ depending on n, k, τ , such that for all $r \in (0, r_0)$, we have

$$\frac{d}{dr}\phi^\gamma(r) \geq -Cr^{\varepsilon-1}, \quad \phi^\gamma(r) \leq C \quad \text{and} \quad \frac{\int_{B_r} |v_r \Delta v_r|}{|H(r, v) + r^{2\gamma}|} \leq Cr^\varepsilon.$$

In particular, $\phi^\gamma(0+, v)$ exists and satisfies $\phi^\gamma(0+, v) \leq \gamma$.

Proof. We prove by iteration that $\phi^\gamma + g^\gamma \leq C(\gamma)$, then the rest follows easily. We start from $\gamma = 0$, which trivially holds. Then the key step is that, if $\phi^\gamma + g^\gamma \leq C_\gamma$, then for $0 < 5\delta < \varepsilon$, we have $\phi^{\gamma+\delta} + g^{\gamma+\delta} \leq C(\gamma, \delta)$. This would finish the proof. Note that ε is decreasing with respect to γ .

In fact, we have

$$\begin{aligned} \phi^{\gamma+\delta}(r) &= \frac{D(r) + (\gamma + \delta)r^{2\gamma+2\delta}}{H(r) + r^{2\gamma+2\delta}} \leq \frac{2}{r^{2\delta}} \frac{D(r) + \gamma r^{2\gamma}}{H(r) + r^{2\gamma}} \leq \frac{C}{r^{2\delta}} \\ g^{\gamma+\delta}(r) &= \frac{\|v_{\theta r}\|_{L^2(B_2 \setminus B_{1/2})}^2}{H(r) + r^{2\gamma+2\delta}} \leq \frac{1}{\gamma^{2\delta}} \frac{\|v_{\theta r}\|_{L^2(B_2 \setminus B_{1/2})}^2}{H(r) + r^{2\gamma}} \leq \frac{C}{r^{2\delta}}. \end{aligned}$$

Now we apply Lemma 5.1 to get

$$\frac{d}{dr}\phi^\gamma(r) \geq -Cr^{\varepsilon-1}(g^{\gamma+\delta}(r) + 1)(\phi^{\gamma+\delta}(r) + 1) \geq -Cr^{\varepsilon-4\delta-1} \geq -Cr^{\delta-1}.$$

Therefore $\phi^{\gamma+\delta} \leq \phi^{\gamma+\delta}(r_0) + Cr_0^\delta \leq C(\gamma, \delta)$. Finally, by Lemma 2.9, we have for $\theta r/2 \leq r_1 < r_2 < 2\theta r$,

$$\frac{H(r_2) + r_2^{2\gamma+2\delta}}{H(r_1) + r_1^{2\gamma+2\delta}} \lesssim \left(\frac{r_2}{r_1}\right)^{C(\gamma, \delta)} \lesssim C(\gamma, \delta).$$

Therefore

$$g^{\gamma+\delta}(r) \lesssim \int_{\theta r/2}^{2\theta r} \frac{H(R, v) dR}{H(r, v) + r^{2\gamma+2\delta}} \leq C(\gamma, \delta). \quad \square$$

The following corollary is much alike Lemma 2.6.

Corollary 5.3. With the same assumptions of Proposition 5.2, we have for $r \in (0, r_0)$,

$$\frac{d}{dr} \log(r^{-2\lambda}(H(r, u - P_k) + r^{2\gamma})) \geq -Cr^{\varepsilon-1},$$

provided $\lambda \leq \phi^\gamma(0+, u - P_k)$, where C depends on n, k, τ .

Proof. Direct computation shows

$$\begin{aligned} \frac{d}{dr} \log(r^{-2\lambda}(H(r, u - P_k) + r^{2\gamma})) &= \frac{2}{r}(\phi^\gamma(r, v) - \lambda) + \frac{2}{r} \frac{\int_{B_1} v_r \Delta v_r}{H(r, v) + r^{2\gamma}} \\ &\geq -C \int_0^r s^{\varepsilon-1} ds - Cr^\varepsilon = -Cr^\varepsilon. \end{aligned} \quad \square$$

5.2 The set Σ^{kth}

We assumed no other relationship between u and P_k in last subsection, except that p_2 is the blow-up of u . In this subsection, we define Σ^{kth} (see Definition 5.4) to be the set of point with an expansion to order k and determine $p_1, \dots, p_k$, and leave p_{k+1} to be free. We study the blow-up of $u - P_k$. We leave until next section to show the “bad set” $\Sigma^{kth} \setminus \Sigma^{(k+1)th}$ is not large.

Definition 5.4. *u is a solution of (1), and $x_0 \in \Sigma_{n-1}$. We define*

$$(1) \quad \Sigma^{2nd} = \Sigma_{n-1}.$$

$$(2) \quad x_0 \in \Sigma^{3rd} \text{ iff as } r \rightarrow 0,$$

$$r^{-3}(u(x_0 + r \cdot) - p_{2,x_0}(r \cdot)) \rightarrow p_{3,x_0},$$

in the sense of weak $H_{loc}^1(\mathbb{R}^n)$ and $C_{loc}^0(\mathbb{R}^n)$, where p_{3,x_0} is a 3-homogeneous harmonic polynomial vanishing on $\{p_{2,x_0} = 0\}$. Thus we have $(p_{2,x_0}, p_{3,x_0}) \in P_3$.

$$(3) \quad x_0 \in \Sigma^{kth} \text{ for } k \geq 4 \text{ iff } x \in \Sigma^{(k-1)th} \text{ and as } r \rightarrow 0,$$

$$r^{-k}(u(x_0 + r \cdot) - P_{k-1,x_0}(r \cdot)) \rightarrow p_{k,x_0},$$

in the sense of weak $H_{loc}^1(\mathbb{R}^n)$ and $C_{loc}^0(\mathbb{R}^n)$, where $P_{k-1,x_0} = P_{k-1}(p_{2,x_0}, \dots, p_{k-1,x_0})$, and p_{k,x_0} is a 3-homogeneous harmonic polynomial vanishing on $\{p_{2,x_0} = 0\}$. Thus we have $(p_{2,x_0}, \dots, p_{k,x_0}) \in P_k$, and P_k refers to this $P_k(p_2, \dots, p_k)$.

When $x_0 = 0$, we simply drop x_0 from p_{k,x_0} and write p_k .

We remark that $r^{-k}(u - P_{k-1}) \rightarrow p_k$ is equivalent to $r^{-k}(u - P_k) \rightarrow 0$, since $P_k = P_{k-1} + p_k + O(|x|^{k+1})$. The latter expression is more useful in proofs.

Lemma 5.5. *Suppose $0 \in \Sigma^{kth}$ and $\gamma \in [k, k+2)$. Then for any q such that $(p_2, \dots, p_{k-1}, q) \in P_k$, we have*

$$\phi^\gamma(0+, u - P_k(p_2, \dots, p_{k-1}, q)) \geq k.$$

Proof. By Lemma 2.9 we have, as $r \rightarrow 0$,

$$r^{2\phi^\gamma(0+, u - P_k) + \delta} \lesssim H(r, u - P_k) + r^{2\gamma}.$$

Recall $\phi^\gamma(0+, u - P_k) \leq \gamma$. If this inequality is strict, then $H(r, u - P_k) \gtrsim r^{2\phi^\gamma(0+, u - P_k) + \delta}$ for any $\delta \ll 1$. On the other hand, since $P_k = P_{k-1} + O(|x|^k)$, and $H(r, u - P_{k-1}) = O(|x|^k)$ (for $r^{-k}(u - P_{k-1}) \rightarrow p_k$ according to the definition of Σ^{kth}),

$$H(r, u - P_k) \lesssim H(r, u - P_{k-1}) + H(r, O(|x|^k)) = O(r^{2k}).$$

Combining these two inequalities, we obtain $\phi^\gamma(0+, u - P_k) \geq k$. □

Corollary 5.3 now implies that $H_k(r, u - P_k) = r^{-2k}H(r, u - P_k)$ is almost monotone.

Lemma 5.6. *Let $0 \in \Sigma^{kth}$, q is a polynomial such that $(p_2, \dots, p_{k-1}, q) \in P_k$, and $|(p_2, \dots, p_{k-1}, q)| \leq \tau$. For $w := u - P_k(p_2, \dots, p_{k-1}, q)$, there exists C, r_0 depending on n, k, τ , such that for all $r \in (0, r_0)$, we have*

$$H_k(r, w) \leq C \quad \text{and} \quad \frac{d}{dr}(H_k(r, w)) \geq -Cr^{\varepsilon-1}.$$

Proof. Choose $\gamma = k + 3/2$, then Lemma 5.5 and Corollary 5.3 implies, for $r \in (0, r_0)$,

$$\frac{d}{dr} \log(r^{-2k}(H(r, w) + r^{2\gamma})) \geq -Cr^{\varepsilon-1}.$$

Therefore $r^{-2k}(H(r, w) + r^{2\gamma})$ is bounded. Thus

$$\frac{d}{dr}(r^{-2k}(H(r, w) + r^{2\gamma})) \geq -Cr^{\varepsilon-1}(r^{-2k}(H(r, w) + r^{2\gamma})) \geq -Cr^{\varepsilon-1}.$$

We are done by observing $\frac{d}{dr}(r^{2(\gamma-k)}) = 1 \ll r^{\varepsilon-1}$. $\square$

Next proposition shows that the map $x \mapsto P_{k,x}$ is continuous. For $k = 2$, the continuity of $x \mapsto p_{2,x}$ was first known by Caffarelli.

Proposition 5.7. *The map $\Sigma^{kth} \ni x \mapsto P_{k,x}$ is continuous. Furthermore, there exists a constant $\tau(n, k)$ such that*

$$\sup\{|(p_{2,x}, \dots, p_{k,x})| \mid x \in \Sigma^{kth} \cap B_{1/2}\} \leq \tau(n, k).$$

Proof. We first prove the existence of bound $\tau(n, k)$ by induction on k . For $k = 2$, $|(p_{2,x})|$ is always a constant. The inductive step follows by applying Lemma 5.6 to $q = 0$ (note that $H_k(r, u - P_k(p_2, \dots, p_{k-1}, 0)) = r^{-2k}\|u - P_{k-1}\|_{L^2(\partial B_1)} \rightarrow \|p_k\|_{L^2(\partial B_1)}$).

Let's prove the continuity at 0. Again, we proceed by induction. When $k = 2$, this was proved by Caffarelli. We provide this proof here for completeness. Suppose $x_l \rightarrow 0$. We have for any $r > 0$

$$|r^{-2}u(rx) - p_{2,x_l}(x)| \leq r^{-2}|u(rx) - u(x_l + rx)| + |r^{-2}u(x_l + rx) - p_{2,x_l}(x)|.$$

Recall $|r^{-2}u(x_l + rx) - p_{2,x_l}(x)| \leq \omega(r)$ for $x \in \partial B_1$ as $r \rightarrow 0$, where $\omega(r)$ is a modulus of continuity (see [1]). Remark 2.6 says $H_2(r, u - p_{2,x_l}) = \int_{\partial B_1} |r^{-2}u_r - p_{2,x_l}|^2$ is increasing, hence

$$\int_{\partial B_1} |p_2 - p_{2,x_l}|^2 = H(0+, u - p_{2,x_l}) \leq H(r, u - p_{2,x_l}) \lesssim r^{-2} \int_{B_1} |u(rx) - u(x_l + rx)|^2 + \omega(r).$$

Let $l \rightarrow \infty$, and since r is arbitrary, we obtain $\int_{\partial B_1} |p_2 - p_{2,x_k}|^2 \rightarrow 0$.

Suppose the continuity is true for $k - 1$, let's consider k . There exists a rotation R_l taking $\{p_{2,x_l} = 0\}$ to $\{p_2 = 0\}$. Since $p_{2,x_l} \rightarrow p_2$, we have $R_l \rightarrow \text{id}$. We apply Lemma 5.6 to $u(x_l + \cdot)$ and $q_l := p_{k,0} \circ R_l$ (it satisfies $(p_{2,x_l}, \dots, p_{k-1,x_l}, q_l) \in P_k$), then

$$r \mapsto Cr^\varepsilon + \int_{\partial B_1} \left| \frac{u(x_l + r \cdot) - P_k(p_{2,x_l}, \dots, p_{k-1,x_l}, q_l)}{r^k} \right|^2$$

in monotone increasing for $r \in (0, r_0)$ and any $l \in \mathbb{N}$. Note that

$$u(x_l + y) - P_k(p_{2,x_l}, \dots, p_{k-1,x_l}, q_l)(y) = u(x_l + y) - P_{k,x_l-1}(y) + q_l(y) + O(|y|^{k+1}).$$

We have

$$\int_{\partial B_1} \left| \frac{u(x_l + rx) - P_k(p_{2,x_l}, \dots, p_{k-1,x_l}, q_l)}{r^k} \right|^2 = \int_{\partial B_1} \left| \frac{u(x_l + r \cdot) - P_{k-1,x_l}(r \cdot)}{r^k} - p_{k,0} \circ R_l + O(r) \right|^2.$$

By definition we have

$$\int_{\partial B_1} \left| \frac{u(r \cdot) - P_{k-1}(r \cdot)}{r^k} - p_k \right|^2 \rightarrow 0, \quad (7)$$

as $r \rightarrow 0$. Therefore

$$\begin{aligned} \int_{\partial B_1} |p_{k,x_l} - p_k \circ R_l|^2 &= \lim_{r \rightarrow 0} \int_{\partial B_1} \left| \frac{u(x_l + r \cdot) - P_{k-1,x_l}(r \cdot)}{r^k} - p_k \circ R_l \right|^2 \\ &\leq Cr^\varepsilon + \int_{\partial B_1} \left| \frac{u(x_l + r \cdot) - P_{k-1,x_l}(r \cdot)}{r^k} - p_{k,0} \circ R_l + O(r) \right|^2 \\ &\leq Cr^\varepsilon + \int_{\partial B_1} \left| \frac{u(x_l + r \cdot) - P_{k-1,x_l}(r \cdot)}{r^k} - p_k \circ R_l \right|^2 \rightarrow 0. \end{aligned}$$

We first choose $r \ll 1$ to make the left hand side of (7) small, then let $l \rightarrow \infty$. By inductive assumption we have $P_{k-1,x_l} \rightarrow P_{k-1}$. Then we are done. $\square$

To end this subsection, we give a more flexible characterization of Σ^{kth} . The following lemma also holds for $k = 2$, if one read it as $r_l^{-2}u_{r_l} \rightarrow q_2$.

Lemma 5.8. *For $k \geq 3$, let $\tilde{\Sigma}^{kth}$ consists of all $x \in \Sigma^{(k-1)th}$ with the following property: there exists $(q_2, \dots, q_k) \in P_k$ and $r_l \rightarrow 0$, such that*

$$r_l^{-k}(u - P_{k-1}(q_2, \dots, q_{k-1}))_{r_l} \rightharpoonup q_k \quad \text{in } H_{loc}^1(\mathbb{R}).$$

Then $\Sigma^{kth} = \tilde{\Sigma}^{kth}$, and $q_j = p_{j,x}$ for $j = 2, \dots, n$.

Proof. We only need to prove $\tilde{\Sigma}^{kth} \subseteq \Sigma^{kth}$. By definition, if $x \in \tilde{\Sigma}^{kth}$, then $x \in \Sigma^{k-1}$. We can assume $x = 0$ without lost of generality. We first show that $q_j = p_j$ for $j = 2, \dots, k-1$. By the compactness of trace operator, we have $H_k(r_l, u - P_k(q_2, \dots, q_k)) \leq C$, so that

$$H_j(r_l, u - P_k(q_2, \dots, q_k)) \rightarrow 0, \quad j = 2, \dots, k-1.$$

The base step $k = 2$ follows by

$$\begin{aligned} H_2(r_l, p_2 - q_2)^{1/2} &\leq H_2(r_l, u - p_2)^{1/2} + H_2(r_l, u - P_2(q_2))^{1/2} \\ &= H_2(r_l, u - p_2)^{1/2} + H_2(r_l, u - P_k(q_2, \dots, q_k))^{1/2} + O(r_l). \end{aligned}$$

Let $l \rightarrow \infty$, we get $p_2 = q_2$. Similarly, if we've proved for $2, \dots, j-1$, then

$$\begin{aligned} H_j(r_l, p_j - q_j)^{1/2} &= H_j(r_l, P_{j-1}(p_2, \dots, p_{j-1}) + p_j - P_{j-1}(q_2, \dots, q_{j-1}) - q_j)^{1/2} \\ &= H_j(r_l, u - P_{j-1}(p_2, \dots, p_{j-1}) - p_j)^{1/2} + H_j(r_l, u - P_k(q_2, \dots, q_k))^{1/2} + O(r_l). \end{aligned}$$

So we obtain $p_j = q_j$.

We temporarily write P_k for $P_k(p_2, \dots, p_{k-1}, q_k)$, and $v := u - P_k$. Then the assumption says $r_l^{-k}v_{r_l} \rightharpoonup 0$ in $H_{loc}^1(\mathbb{R}^n)$. We'll show that this convergence is strong and locally uniformly along $r \rightarrow 0$. This would imply $0 \in \Sigma^{kth}$.

Fix $\gamma = k + 3/2$ and $R > 0$, then for $r \ll 1$,

$$r^{-2k} \int_{B_R} |\nabla v_r|^2 \leq r^{-2k} \phi^\gamma(Rr, v) (H(Rr, v) + (Rr)^{2\gamma}) \leq Cr^{-2k} (H(Rr, v) + (Rr)^{2\gamma}).$$

Lemma 5.3 says the logarithm of the right hand side is almost monotone, therefore

$$\begin{aligned} \limsup_{r \rightarrow 0} \log \left(r^{-2k} \int_{B_R} |\nabla v_r|^2 \right) &\leq \log C + \lim_{s \rightarrow 0} \log (s^{-2k} (H(s, v) + s^{2\gamma})) \\ &\leq \log C + \lim_{l \rightarrow \infty} \log (r_l^{-2k} (H(r_l, v) + r_l^{2\gamma})) = -\infty, \end{aligned}$$

where we used $r_l^{-k} v_{r_l} \rightarrow 0$. Therefore $\|r^{-k} \nabla v_k\|_{L^2(B_R)} \rightarrow 0$, this implies $r^{-k} v_k \rightarrow 0$ in $H^1(B_R)$. Finally, the uniform convergence follows by Lemma 4.7,

$$\|v_r\|_{L^\infty(B_R)} \leq C \|v_{Rr}\|_{L^2(B_2 \setminus B_1)} + C(Rr)^{k+2} \leq CH(Rr, v)^{1/2} + C(Rr)^{k+2} = o(r^k). \quad \square$$

5.3 Blow-up analysis

In this subsection we investigate for $x \in \Sigma^{kth}$ the possible blow-up of

$$\tilde{v}_r := \frac{(u(x + \cdot) - P_{k,x})_r}{H(r, u(x + \cdot) - P_{k,x})^{1/2}},$$

which is very similar to Proposition 3.2 when we studied second order blow-up.

Definition 5.9. For $k \geq 2$, define the k -th frequency $\lambda_k : \Sigma^{kth} \rightarrow [k, k+2]$ by

$$\lambda_k(x) := \sup\{\phi^\gamma(0+, u(x + \cdot) - P_{k,x}) \mid \gamma \in [k, k+2]\}.$$

When $x = 0$, we write λ_k for $\lambda_k(0)$.

We comment that λ indeed belongs to $[k, k+2]$. On the one hand, Proposition 5.2 says $\phi^\gamma(0+, u(x + \cdot) - P_{k,x}) \leq \gamma < k+2$. On the other hand, Lemma 5.5 says $\phi^\gamma(0+, u(x + \cdot) - P_{k,x}) \geq k$ for all $\lambda \in [k, k+2]$. In fact, we shall show $\phi^\gamma(0+, u - P_k) = \min\{\lambda_k, \gamma\}$, which justifies the name “truncated frequency”.

Lemma 5.10. Let $k \geq 2$ and $0 \in \Sigma_k$, we have $\phi^\gamma(0, u - P_k) = \min\{\lambda_k, \gamma\}$. Moreover, if $\lambda_k < k+2$, then

$$\lambda_k = \lim_{r \rightarrow 0} \phi(r, u - P_k),$$

which is the non-truncated Almgren frequency⁷.

Proof. We make two observations:

- Suppose $\phi^\gamma(0+, u - P_k) < \gamma$ (this happens if $\gamma > \lambda_k$), then by Lemma 2.9, we have

$$r^{2\phi^\gamma(0+, u - P_k)} \ll r^{2\gamma - \delta} \lesssim H(r, u - P_k) + r^{2\gamma},$$

so that $H(r, u - P_k) \gtrsim r^{2\gamma - \delta}$. In other words, $r^{2\gamma}/H(r, u - P_k) = o(1)$. Therefore, for any $\tilde{\gamma} \in [\gamma, k+2]$,

$$\phi^{\tilde{\gamma}}(0+, u - P_k) = \lim_{r \rightarrow 0} \frac{\phi(r, u - P_k) + o(1)}{1 + o(1)} = \lim_{r \rightarrow 0} \phi(r, u - P_k).$$

⁷Even though $\phi(r, u - P_k)$ need not be monotone, we still have this limit exists, and call it Almgren frequency.

- Suppose $\phi^\gamma(0+, u - P_k) = \gamma$, then $H(r, u - P_k) + r^{2\gamma} \lesssim r^{2\phi^\gamma(0+, u - P_k)} = r^{2\gamma}$. The limit

$$\phi^\gamma(0+, u - P_k) = \lim_{r \rightarrow 0} \frac{r^{-2\gamma} D(r, u - P_k) + \gamma}{O(1) + 1}$$

exists, implying $D(r, u - P_k) \lesssim r^{2\gamma}$. Therefore, for any $\tilde{\gamma} \in [k, \gamma)$,

$$\phi^{\tilde{\gamma}}(0+, u - P_k) = \lim_{r \rightarrow 0} \frac{o(1) + \tilde{\gamma}}{o(1) + 1} = \tilde{\gamma}.$$

If $\lambda_k = k + 2$, we must have $\phi^\gamma(0+, u - P_k) = \gamma$ for all $\gamma \in [k, k + 2)$. If $\lambda_k < k + 2$, then $\lambda_k \geq \phi(0+, u - P_k)$ for there exists $\gamma > \lambda_k$. We must have $\phi^{\lambda_k}(0+, u - P_k) = \lambda_k$, for otherwise

$$\lambda_k < \phi^{\lambda_k}(0+, u - P_k) = \phi(0+, u - P_k).$$

Finally, since $r^{2\phi(0+, u - P_k) - \delta} \lesssim H(r, u - P_k) \lesssim r^{2\lambda_k}$ for all $\delta \ll 1$, we obtain $\lambda_k = \phi(0+, u - P_k)$. Now we are done. $\square$

Lemma 5.11. *Suppose $0 \in \Sigma^{kth}$ and $\lambda_k > k + 1$, then $0 \in \Sigma^{(k+1)th}$ with $p_{k+1} = 0$.*

Proof. Let $v := u - P_k$. Pick $\gamma \in k + 3/2$, then $\phi^\gamma(0+, u - P_k) = \min\{\lambda_k, \gamma\} > k + 1$. We have

$$H(r, v) = o(r^{k+1}) \quad \text{and} \quad D(r, v) \leq C(H(r, v) + r^{2\gamma}) = o(r^{k+1}).$$

Therefore $r^{-k-1}v_r \rightarrow 0$ in $H_{loc}^1(\mathbb{R}^n)$, which means $0 \in \Sigma^{(k+1)th}$ and $p_{k+1} = 0$ by Lemma 5.8. $\square$

Now we state the main result of this subsection. We allow the points x_l to vary along radius r_l for later use. But the most important case is $x_l = 0$

Proposition 5.12. *Let $0 \in \Sigma^{kth}$, $\lambda_k \leq k + 1$, $r_l \rightarrow 0$ and $x_l \in \Sigma^{kth} \cap B_{r_l}$. We set $v_l := u(x_l + \cdot) - P_{k, x_l}$ and suppose $\lambda_k(x_l) \rightarrow \lambda_k$. Consider the sequence⁸*

$$\tilde{v}_l := \frac{v_l(r_l \cdot)}{H(r_l, v_l)^{1/2}},$$

then

- (1) $\tilde{v}_l$ is bounded in $H_{loc}^1(\mathbb{R}^n)$ and $C_{loc}^{0, \frac{1}{n+1}}(\mathbb{R}^n)$.
- (2) Up to a subsequence, $\tilde{v}_l \rightarrow q$ weakly in $H_{loc}^1(\mathbb{R}^n)$. This convergence is in fact strong and uniform. Moreover, $q \neq 0$ is a λ_k -homogeneous solution to the thin obstacle problem (2) with obstacle 0 on $\{p_2 = 0\}$. Finally, if $\lambda_k < k + 1$, then q is even with respect to $\{p_2 = 0\}$.

We start with a lemma which will be used for proving Hölder estimate in (1). Note that we can't directly obtain a Lipschitz bound for $\tilde{v}_l$, even though we have proved Lipschitz estimate in Section 4, due to the extra exponent β appeared in Proposition 4.6(2). We only have Lipschitz estimate for $\tilde{v}_l$ in the first $n - 1$ components.

⁸One should pay attention to the notation here, which might be confusing. v_l depends on x_l , and is not rescaled. But $\tilde{v}_l$ is rescaled by r_l . We need to use a more cumbersome name $(v_l)_{r_l}$ for the rescaled function.

Lemma 5.13. *Let $w \in C^{0,1}(\overline{B_2})$, then*

$$\|w\|_{C^{0, \frac{1}{n+1}}(B_1)} \leq C(n) \left(\sum_{1 \leq j \leq n-1} \|\partial_j w\|_{L^\infty(B_2)} + \|\partial_n w\|_{L^2(B_2)} + \|w\|_{L^\infty(B_2)} \right).$$

Proof. By homogeneity we can assume the right hand side is 1, and we only need to care for the n th component. For $r < 1$, consider $A_r := B'_{r^{1/n+1}} \times [0, r] \subseteq B_1 \subseteq B_2$, where B'_r denotes the ball in $\mathbb{R}^{n-1}$. We can find some $z' \in B'_{r^\theta}$ such that

$$\int_0^r |\partial_n w(z', s)|^2 dx \leq C(n) r^{-\frac{n-1}{n+1}} \|\partial_n w\|_{L^2(A_r)}^2 \leq C(n) r^{-\frac{n-1}{n+1}}.$$

Finally, we have

$$\begin{aligned} |w(0, 0) - w(0, r)| &\leq |w(0) - w(z', 0)| + |w(z', 0) - w(z', r)| - |w(z', r) - w(0, r)| \\ &\leq C(n) \left(|z'| + \int_0^r |\partial_n w(z', s)| ds \right) \\ &\leq C(n) \left(r^{\frac{1}{n+1}} + r^{1/2} \left(\int_0^r |\partial_n w(z', x)|^2 ds \right)^{1/2} \right) \\ &\leq C(n) \left(r^{\frac{1}{n+1}} + r^{\frac{1}{2}(1-\frac{n-1}{n+1})} \right) = C(n) r^{\frac{1}{n+1}}. \end{aligned}$$

From this one easily finishes the proof. $\square$

Proof of Proposition 5.12. Set $\delta := \frac{\varepsilon}{100}$ where $\varepsilon(n, k)$ is the same as Lemma 5.6. Without loss of generality assume l is so large that $x_l \in B_{1/2}$ and $\lambda_k(x_l) \leq \lambda_k + \delta$. Within this proof we fix $\gamma := k + 7/4$, so that $\lambda_k(x_l) = \phi^\gamma(0^+, v_l)$ for all l .

Step 1. We claim that there are $\varepsilon, r_0 \in (0, 1/2)$ and $C_0, c_0 > 0$ all independent on l , and l_0 such that if we set

$$f_{x_l}(r) := \phi^\gamma(r, v_l) + C_0 r^\varepsilon, \quad h_{x_l}(r) := r^{-2k} H(r, v_l) + C_0 r^\varepsilon$$

then we have

- (a) f_{x_l}, h_{x_l} are continuous and increasing on $[0, r_0]$ and converge uniformly in this interval to f_0, h_0 respectively. Furthermore $h_{x_l}(0^+) = 0$ identically.
- (b) We have $f_{x_l}(r) \leq \lambda_k + 2\delta$ and $H(r, v_l) \geq c_0 r^{2\lambda_k + 5\delta}$ for all $r \in [0, r_0]$ and all $l > l_0$.

The fact that for each l both f_{x_l} and h_{x_l} are increasing is consequence of Lemma 5.2 and Lemma 5.6, respectively. By Lemma 5.7 we can choose $\tau := \tau(n, k)$ so that r_0 and C_0 can be taken uniform in l . Since $x_l \in \Sigma^{kth}$ we find that $h_{x_l}(0^+) = 0$, furthermore by assumption $f_{x_l}(0^+) = \lambda_k(x_l) \rightarrow \lambda_k = f_0(0^+)$, thus $f_{x_l} \rightarrow f_0$ and $h_{x_l} \rightarrow h_0$ pointwise. As they are monotone and the limit functions are continuous, the convergence must be uniform, thus (a) is proved. We turn to (b): possibly taking a smaller r_0 , we have that $f_0 \leq \lambda_k + \delta$ in $[0, r_0]$ thus by uniform convergence there is l_0 such that $f_{x_l} \leq \lambda_k + 2\delta$. Now the last statement follows from Lemma 2.9.

We will use many times through the proof that

$$c_0 r^{2\gamma} \leq H(r, v_l) r^{\frac{1}{2}} \quad \text{in } [0, r_0] \tag{8}$$

uniformly in $l > l_0$. This is a direct consequence of (b) and the fact that $2\gamma - \frac{1}{2} > 2\lambda_k + 5\delta$.

Step 2. We prove (1). Fix some $R > 1$ and for $Rr_l < r_0$,

$$D(R, \tilde{v}_l) \leq \phi^\gamma(Rr_l, v_l) \frac{H(Rr_l, v_l) + (Rr_l)^{2\gamma}}{H(r_l, v_l)} \leq C_R \frac{H(Rr_l, v_l)}{H(r_l, v_l)} + C_R \frac{r_l^{2\gamma}}{H(r_l, v_l)} \leq C_R + o(1)$$

where we used (8). Thus $\|\nabla \tilde{v}_l\|_{L^2(B_R)}$ is bounded in l .

Now we want to combine the uniform L^2 -bound on $\nabla \tilde{v}_l$ and the Lipschitz estimate on $\nabla_{x'} \tilde{v}_r$ to entail uniform Hölder bounds. Fix some l and choose coordinates such that $p_{2,x_l} = \frac{1}{2}x_n^2$, by Proposition 4.6 we have for $2\theta Rr_l < r_0$ and $j \neq n$

$$\|\partial_j \tilde{v}_l\|_{L^\infty(B_R)} \leq C \|\tilde{v}_l\|_{L^2(B_{2\theta R} \setminus B_{\theta R/2})} + C \frac{r^{k+2}}{H(r_l, v_l)^{\frac{1}{2}}}$$

the second term is $o(1)$ in l again by (8), while for the first we employ again Lemma 2.9:

$$\|v_l(r_l \cdot)\|_{L^2(B_{2\theta R} \setminus B_{\theta R/2})}^2 = \int_{\theta R/2}^{2\theta R} H(sr_l, v_l) s^{n-1} ds \leq C_R H(r_l, v_l).$$

Hence, $\|\partial_j \tilde{v}_l\|_{L^\infty(B_R)} \leq C_R$ for all $j \neq n$. Similarly, we find $\|\tilde{v}_l\|_{L^\infty(C_R)} \leq C_R$ from Lemma 4.7. By Lemma 5.13, this gives the Hölder bound: $\|\tilde{v}_l\|_{C^{0, \frac{1}{n+1}}(B_{R/2})} \leq C_R$. This concludes the proof of (1).

Step 3. We turn to (2). By (1) we have $\tilde{v}_r \rightarrow q$ in $H_{\text{loc}}^1(\mathbb{R}^n)$. Let's prove that q solves the thin obstacle problem. Since $\Delta v_l = -\chi_{\{u(x_l + \cdot) = 0\}} \leq 0$, we have that $\Delta q \leq 0$ weakly in $\mathbb{R}^n$. Furthermore, integrating by parts with some cut-off function $\chi_{B_R} \leq \psi \leq \chi_{B_{2R}}$ leads to

$$\int_{B_R} |\Delta \tilde{v}_l| \leq - \int_{\mathbb{R}^n} \Delta \tilde{v}_l \psi \leq C_R \|\tilde{v}_l\|_{L^2(B_{2R} \setminus B_R)} \leq C_R.$$

Hence $\Delta \tilde{v}_l \xrightarrow{*} \Delta q$ as a Radon measure. Since $\tilde{v}_l \rightarrow q$ locally uniformly, we have $\tilde{v}_l \Delta \tilde{v}_l \xrightarrow{*} q \Delta q$.

For $Rr_l < r_0$, by Proposition 5.2,

$$\int_{B_R} |\tilde{v}_l \Delta \tilde{v}_l| \frac{H(r_l, v_l)}{H(Rr_l, v_l) + (Rr_l)^{2\gamma}} \leq C_R r_l^\varepsilon = o(1).$$

The constants are independent on l as we can choose a uniform $\tau = \tau(n, k)$ for all $x_l \in \Sigma^{kth} \cap B_{1/2}$ by Proposition 5.7. Sending $l \uparrow \infty$ we get $q \Delta q = 0$, where we used $H(r_l, v_l) \gg r_l^{2\gamma}$, so as $l \rightarrow \infty$,

$$\frac{H(r_l, v_l)}{H(Rr_l, v_l) + (Rr_l)^{2\gamma}} \gtrsim \frac{H(r_l, v_l) + r_l^{2\gamma}}{H(Rr_l, v_l) + (Rr_l)^{2\gamma}} \gtrsim C_R.$$

In order to show $\Delta q = 0$ outside $\{p_2 = 0\}$, we first exploit Lemma 3.10 to find

$$B_{R/2} \cap \text{supp}(\Delta \tilde{v}_l) \subseteq \{\text{dist}(\{p_{2,x_l} = 0\}, \cdot) \leq C_R r_l^{\alpha_\circ}\}$$

as $p_{2,x_l} \rightarrow p_2$ and R can be taken arbitrarily large we deduce that $\Delta q \subseteq$ is supported in $\{p_2 = 0\}$.

It remains to show that q is non-negative on $\{p_2 = 0\}$. Up to a rotation we can assume $p_{2,0} = \frac{1}{2}x_n^2$. For any $x_* \in \{x_n = 0\}$, we claim that it's possible to find a sequence $(y_l)_{l \in \mathbb{N}}$ such that

$$y_l \in \{\mathcal{A}_{k,x_l}(r_l \cdot) = 0\}, \quad y_l \rightarrow x_*.$$

Then by locally uniform convergence,

$$q(x_*) = \lim_l \tilde{v}_{r_l}(y_l) = \lim_l \frac{u(r_l y_l) - \frac{1}{2} \mathcal{A}_{k,x_l}^2(r_l y_l) + O(r_l^{k+2})}{H(r_l, v_l)^{1/2}} \geq 0.$$

To check our claim, set $y_l := \Phi_l(x_*)$ where $\Phi_l \in C^\infty(B_{R_l})$ are the inverse functions of $\Psi_l : x \mapsto (x', r_l^{-1}(\mathcal{A}_{k,x_l})_{r_l})$. Notice that $\Psi_l \rightarrow \text{id}$ in C_{loc}^1 and that $R_l \uparrow +\infty$ as $l \rightarrow \infty$. So $\Phi_l \rightarrow \text{id}$, and $x_* \in B_{R_l}$ eventually, thus $y_l \rightarrow x_*$. Therefore we have proved that q is a global solution of the thin obstacle problem.

Step 4. We show that $\tilde{v}_l \rightarrow q$ in $H_{\text{loc}}^1(\mathbb{R}^n)$ and that q is λ_k -homogeneous. Fix any $\eta \in C_c^\infty(\mathbb{R}^n)$ and exploit as before that $\|\tilde{v}_{r_l} \Delta \tilde{v}_{r_l}\|_{L^1(B_R)} \rightarrow 0$ and integrate by parts in $\mathbb{R}^n$

$$\begin{aligned} \int |\nabla(\eta \tilde{v}_l)|^2 &= - \int \eta \tilde{v}_l \Delta(\eta \tilde{v}_l) \\ &= - \int (\eta \tilde{v}_l^2 \Delta \eta + 2\eta \tilde{v}_l \nabla \eta \cdot \nabla \tilde{v}_l + \eta^2 \tilde{v}_l \Delta \tilde{v}_l) \\ &\leq - \int (\eta \Delta \eta \tilde{v}_l^2 + 2\eta \tilde{v}_l \nabla \eta \cdot \nabla \tilde{v}_l) + C(\eta) \|\tilde{v}_l \Delta \tilde{v}_l\|_{L^1(B_R)} \end{aligned}$$

Taking the upper limit and using $\nabla \tilde{v}_l \rightharpoonup \nabla q$ in $L_{\text{loc}}^2(\mathbb{R}^n)$ and $\tilde{v}_l \rightarrow q$ in $C_{\text{loc}}^0(\mathbb{R}^n)$, we get

$$\limsup_l \int |\nabla(\eta \tilde{v}_l)|^2 \leq - \int (\eta \Delta \eta q^2 + 2\eta q \nabla \eta \cdot \nabla q + \eta^2 q \Delta q) = \int |\nabla(\eta q)|^2$$

where we used $q \Delta q = 0$. By weak lower-semicontinuity we always have the converse inequality thus $\nabla(\eta \tilde{v}_l) \rightarrow \nabla(\eta q)$ strongly in $L^2(\mathbb{R}^n)$.

In particular, for every $R > 0$, together with (a) in Step 1 we find,

$$\begin{aligned} \phi(R, q) &= \lim_l \phi(R, \tilde{v}_l) = \lim_l \phi(Rr_l, v_l) = \lim_l \phi^\gamma(Rr_l, v_l) \\ &= (Rr_l, v_l) = \lim_l f_{x_l}(Rr_l) = f_0(0^+) = \lambda_k. \end{aligned}$$

This implies q is λ_k -homogeneous.

Step 5. We finally prove that for $\lambda_k < k+1$ we have $q^{\text{odd}} = 0$. Notice that by q^{odd} is harmonic thus has integral homogeneity, hence the only nontrivial case is when $\lambda_k = k$. We need to show that q is orthogonal in $L^2(\partial B_1)$ to every k -homogeneous harmonic polynomial P vanishing on $\{p_2 = 0\}$. Fix such P and apply Lemma 5.6 with

$$w_l := u(x_l + \cdot) - P_k(p_{2,x_l}, \dots, p_{k-1,x_l}, p_{k,x_l} - P \circ R_l),$$

where R_l are rotations sending $\{p_{2,x_l} = 0\}$ to $\{p_2 = 0\}$ and $R_l \rightarrow \text{id}$. Thus we have with constants uniform in l (possibly depends on P)

$$\begin{aligned} r^{-2k} H(r, w_l) + Cr^\varepsilon &= Cr^\varepsilon + \int_{\partial B_1} \left(\frac{v_l(r \cdot)}{r^k} + P \circ R_l + O(r) \right)^2 \\ &\geq \lim_{r \rightarrow 0} (r^{-2k} H(r, w_l) + Cr^\varepsilon) = \int_{\partial B_1} P^2 \end{aligned}$$

Divide by the sequence $\varepsilon_l := (H(r_l, v_l) r_l^{-2k})^{1/2}$, which by Step 1 (b) satisfies (recall $\lambda_k = k$)

$$r_l^{5\delta} \leq \varepsilon_l^2 \leq h_{x_l}(r_l) = o(1).$$

We compute the squares and rearrange the terms to get:

$$\int_{\partial B_1} \tilde{v}_l^2 \varepsilon_l + 2 \int_{\partial B_1} \tilde{v}_l P \circ R_l \geq -C \frac{r_l^\varepsilon}{\varepsilon_l} \geq -C r_l^{\varepsilon - 5\delta/2}$$

Since $\delta \leq \varepsilon/100$ we can send $l \rightarrow \infty$ and get

$$\int_{\partial B_1} qP \geq 0$$

Therefore $q^{\text{odd}} = 0$. □

6 Estimating the size of $\Sigma^{kth} \setminus \Sigma^{(k+1)th}$

The goal of this section is to show for $k \geq 2$

$$\dim_{\mathcal{H}}(\Sigma^{kth} \setminus \Sigma^{(k+1)th}) \leq n - 2,$$

and $\Sigma^{kth} \setminus \Sigma^{(k+1)th}$ is countable if $n = 2$. Recall Lemma 5.11 shows if $0 \in \Sigma^{kth} \setminus \Sigma^{(k+1)th}$, then $\lambda_k \leq k + 1$. Also

$$r^{-k-1}(u - P_k)_r = \frac{H(r, u - P_k)^{1/2}}{r^{k+1}} \frac{(u - P_k)_r}{H(r, u - P_k)^{1/2}} = O(r^{\lambda_k - (k+1)}) \tilde{v}_r,$$

either $\lambda_k < k + 1$, or $\lambda_k = k + 1$ and every accumulation point q of $\tilde{v}_r$ as $r \rightarrow 0$ satisfies $q^{\text{even}} \neq 0$. Let

$$\Sigma^{>k} := \{x \in \Sigma^{kth} \mid \lambda_k > k\} \quad \text{and} \quad \Sigma^{\geq k+1} := \{x \in \Sigma^{kth} \mid \lambda_k \geq k + 1\},$$

One easily verifies

$$\Sigma^{(k+1)th} \subseteq \Sigma^{\geq k+1} \subseteq \Sigma^{>k} \subseteq \Sigma^{kth}.$$

We divide $\Sigma^{kth} \setminus \Sigma^{(k+1)th}$ into three parts, $\Sigma^{kth} \setminus \Sigma^{>k}$, $\Sigma^{>k} \setminus \Sigma^{\geq k+1}$ and $\Sigma^{\geq k+1} \setminus \Sigma^{(k+1)th}$, and investigate them respectively. We shall see in the proofs that the parity of k plays a role, because the homogeneity of the $2D$ thin obstacle problem consists of

$$\mathbb{N} \cup \{2m - \frac{1}{2} \mid m \in \mathbb{N}\}.$$

6.1 Dimension reduction tools

In this subsection, we state and prove several lemmas that would be used in dimension reduction arguments later. They are purely geometric-measure-theoretic issues, and has nothing to do with the obstacle problem.

The following lemma follows from a result of A. Naber and D. Valtoria in [5] on the estimate of the critical set of Harmonic functions.

Lemma 6.1. *Let $\beta > n - 2$ be a real number and $k \geq 3$ be odd, then there exists $\varepsilon_0(n) \ll 1$ such that for all $\varepsilon < \varepsilon_0(n, \beta)$ the following holds. If the set $E \subseteq \mathbb{R}^n$ satisfies*

$$E \subseteq \overline{B_r(x)} \cap (\Sigma(q) + \overline{B_{\varepsilon r}(x)}),$$

for some $r \in (0, 1)$, $x \in \mathbb{R}^n$ and $0 \neq q \in S_k^{\text{even}}(L)$. Then E can be covered by $\varepsilon^{-\beta}$ balls of radius εr centered in E .

Proof. By translation scaling and rotation we may assume $x = 0$, $r = 1$ and $L = \{x_n = 0\}$. Lemma 3.1 promises q vanishes on L (this is why we are assuming k to be odd). Let Q be the odd extension of $q|_{\{x_n > 0\}}$ to $\mathbb{R}^n$. Clearly $\Sigma(q) \subseteq \Sigma(Q)$, where $\Sigma(Q) := \{Q = |\nabla Q| = 0\}$. Hence

$$E \subseteq B_1 \cap \overline{B_\varepsilon(\Sigma(Q))}.$$

Now we can apply [5, Theorem 1.1] to obtain

$$|B_2 \cap B_t(\Sigma(u))| \leq C(n)t^2,$$

for all $t \in (0, 1)$. This volume, also known as the Minkowski content, is equivalent to covering a set of radius t . More precisely, consider a maximal disjoint subset $\{x_i \mid i \in I\}$ of E with $|x_i - x_j| \geq \varepsilon/2$ for any $i \neq j$. Then $E \subseteq \cup_{i \in I} B_\varepsilon(x_i)$. We can estimate $\#I$ by noticing $B_{\varepsilon/4}(x_i)$ are disjoint and $B_{\varepsilon/4} \subseteq B_2 \cap B_{2\varepsilon}(\Sigma(Q))$, thus

$$\omega_n(\varepsilon/4)^n \#I = |\cup_{i \in I} B_{\varepsilon/4}(x_i)| \leq |B_2 \cap B_{2\varepsilon}(\Sigma(Q))| \leq C(n)\varepsilon^2.$$

Finally, as $\beta > n - 2$, we have $\#I \leq C(n)\varepsilon^{2-n} \leq \varepsilon^{-\beta}$ for $\varepsilon \ll 1$. $\square$

Lemma 6.2. *Let $k \geq 3$ be odd, $E \subseteq \mathbb{R}^n$ and $\tau: E \rightarrow \mathbb{R}$ be a lower-semicontinuous function. Assume for any $\varepsilon > 0$ and $x \in E$, there exists $\rho = \rho(x, \varepsilon) > 0$ such that, for all $r \in (0, \rho)$, we have*

$$E \cap \overline{B_r(x)} \cap \tau^{-1}([\tau(x), \infty)) \subseteq \overline{B_{\varepsilon r}(x)} + \Sigma(q),$$

where $0 \neq q \in S_k^{\text{even}}(L)$. Then $\dim_{\mathcal{H}}(E) \leq n - 2$.

Proof. Fix $\beta > n - 2$ and $\bar{\beta} \in (n - 2, \beta)$, we need to show $\mathcal{H}^\beta(E) = 0$. Let $\varepsilon \ll 1$ to be determined later. Decompose E as $E = \cup_k E_k$, where

$$E_k := \{x \in E \mid \rho(x) > 1/k\}.$$

Then it suffices to show each E_k satisfies $\mathcal{H}^\beta(E_k) = 0$. And due to the inner regularity of $\mathcal{H}^\beta$, we only need to prove $\mathcal{H}^\beta(K) = 0$ for any compact set $K \subseteq E_k$.

For any $r \in (0, 1/(2k))$ and $x \in K$, τ attains a minimum in $K \cap \overline{B_r(x)}$ at some point $\bar{x} \in K \cap \overline{B_r(x)}$ due to lower-semicontinuity. Therefore

$$K \cap \overline{B_r(x)} \subseteq E \cap \overline{B_{2r}(\bar{x})} \cap \tau^{-1}([\tau(\bar{x}), \infty)) \subseteq \overline{B_{2\varepsilon r}(x)} + \Sigma(q),$$

for some $q \in S_k^{\text{even}}(L)$. By Lemma 6.1, we can cover $K \cap \overline{B_r(x)}$ by $\varepsilon^{-\bar{\beta}}$ balls of radius εr if $\varepsilon \ll 1$.

The rest is easy. We initially choose a finite covering of M closed balls of radius $r_0 = 1/(4k)$ centered in K . Then we can replace each $K \cap \overline{B_{r_0}(x)}$ by $\varepsilon^{-\bar{\beta}}$ balls of radius $r_1 = \varepsilon r_0$. In general, at the k th step, we can replace each $K \cap \overline{B_{r_k}(x)}$ by $\varepsilon^{-\bar{\beta}}$ balls of radius $r_k = \varepsilon^k r_0$. In total there are $\varepsilon^{-k\bar{\beta}} M$ balls. Hence

$$\mathcal{H}_\infty^\beta(K) \leq \omega_\beta \varepsilon^{k\beta} r_0^k \times \varepsilon^{-k\bar{\beta}} M = C \varepsilon^{k(\beta - \bar{\beta})} \rightarrow 0. \quad \square$$

We point out that either or not taking closure doesn't matter at all, as one can change ε a little bit. Here we use closed balls only to simplify the proof. There's another lemma in the same spirit. The proof is omitted since it's almost the same (and simpler). The main difference is that now we should cut E up into the sets $E_{k,j} := \{x \in E \mid \rho(x, \varepsilon) \geq 1/k, f(x) \in (j/k, j+1/k]\}$.

Lemma 6.3. *Let $E \subseteq \mathbb{R}^n$, $f: E \rightarrow \mathbb{R}$ and $m \in \{1, \dots, n\}$. Assume for any $\varepsilon > 0$ and $x \in E$, there exists $\rho(x, \varepsilon) > 0$, such that for all $r \in (0, \rho)$, we have*

$$E \cap \overline{B_r(x)} \cap f^{-1}([f(x) - \rho, f(x) + \rho]) \subseteq B_{\varepsilon r}(\Pi_{x,r}^m),$$

where $\Pi_{x,r}^m$ is a m -dimensional plane of $\mathbb{R}^n$, might depends on x, r . Then $\dim_{\mathcal{H}}(E) \leq m$.

6.2 The size of $\Sigma^{kth} \setminus \Sigma^{>k}$

We start with estimating $\Sigma^{kth} \setminus \Sigma^{>k}$. We first show, for even k that $\Sigma^{kth} \setminus \Sigma^{>k} = \emptyset$, from the following monotonicity formula.

Lemma 6.4. *Let $0 \in \Sigma^{kth}$ and $v := u - P_k$. Suppose P is any k -homogeneous harmonic polynomial satisfying $P \geq 0$ on $\{p_2 = 0\}$, then there exists $r_0 = r_0(n, k)$, such that for all $r \in (0, r_0)$, we have*

$$\frac{d}{dr} \left(r^{-k} \int_{\partial B_1} v_r P \right) \leq C r^{\varepsilon-1},$$

where the constants C depends on $n, k, \|P\|_{L^2(\partial B_1)}$ and $\varepsilon > 0$ is dimensional.

Proof. By direct computation, as well as $\Delta P = 0$ and $x \cdot \nabla P = kP$, we have

$$\frac{d}{dr} \int_{\partial B_1} v_r P = \frac{1}{r} \int_{\partial B_1} x \cdot \nabla v_r P = \frac{1}{r} \left(k \int_{\partial B_1} v_r P + \int_{B_1} P \Delta v_r \right).$$

As $\Delta v_r = -r^2 \chi_{\{u_r=0\}}$, we can write this as

$$\frac{d}{dr} \left(r^{-k} \int_{\partial B_1} v_r P \right) = -r^{1-k} \int_{B_1 \cap \{u_r=0\}} P.$$

Assume $p_2 = \frac{1}{2}x_n^2$. Since $P \geq 0$ on $\{p_2 = 0\}$, we have $-P \leq C|x_n|$ in B_1 . In particular, in $B_1 \cap \{u_r = 0\}$, we have $-P \leq C r^{\alpha_0}$. Since $\|v_r\|_{L^2(B_1)} = H(r, v)^{1/2} \|\tilde{v}_r\|_{L^2(B_1)} = O(r^k)$, together with Proposition 4.6, we find

$$B_1 \cap \{u_r = 0\} \subseteq \{x \in B_1 \mid r|\partial_r P_k|(rx) = r^2|x_n + O(r)| \leq C r^{k(1-\beta)}\},$$

where $\beta \in (0, \frac{\alpha_0^2}{k+2})$. If we fix $\beta \ll 1$, then for $r \ll 1$ we obtain

$$\frac{d}{dr} \left(r^{-k} \int_{\partial B_1} v_r P \right) \leq C r^{1-k+\alpha_0} |B_1 \cap \{u_r = 0\}| \leq C r^{\alpha_0-k\beta-2} =: C r^{\varepsilon-1}. \quad \square$$

Corollary 6.5. *For even $k \geq 2$, $\Sigma^{kth} \setminus \Sigma^{>k} = \emptyset$.*

Proof. Assume on the contrary that $0 \in \Sigma^{kth} \setminus \Sigma^{>k}$, in other words $\lambda_k = k$. Then by Proposition 5.12, any accumulation point q of $\tilde{v}_r$ is a non-zero solution of the thin obstacle problem which lies in $S_k^{\text{even}}(\{p_2 = 0\})$, which is in fact harmonic and $q \geq 0$ on $\{p_2 = 0\}$ by Lemma 3.1. From $0 \in \Sigma^{kth}$ we know $r^{-k}v_r \rightarrow 0$. Combining this with Lemma 6.4, we get

$$r^{-k} \int_{\partial B_1} v_r q \leq Cr^\varepsilon.$$

Rewrite this as

$$\int_{\partial B_1} \tilde{v}_{r_l} q \leq \frac{Cr_l^{k+\varepsilon}}{H(r_l, u - P_k)^{1/2}}.$$

As $l \rightarrow \infty$, right hand side vanishes, and we find $\int_{\partial B_1} q^2 \leq 0$. Contradictory! $\square$

Things are a little more involved for odd k . In [2, Proposition 5.8], it was a very dedicate proof that showed $\Sigma^{3rd} \setminus \Sigma^{>3} = \emptyset$. We only prove here a weaker result.

Lemma 6.6. *Let $k \geq 3$ be odd. For any $x \in \Sigma^{kth} \setminus \Sigma^{>k}$ and $\varepsilon > 0$, there exists $\rho = \rho(\varepsilon, x)$, such that for all $r \in (0, \rho)$, there exists $q \in S_k^{\text{even}}(\{p_{2,x} = 0\})$ such that*

$$\Sigma(u) \cap B_r(x) \subseteq \Sigma(q) + B_{\varepsilon r}(x),$$

where $\Sigma(q) := \{q = |\nabla q| = 0\} \cap \{p_2 = 0\}$ and the sum refers to the Minkowski sum of two sets.

Proof. We may suppose $x = 0$, $p_2 = \frac{1}{2}x_n^2$, and argue by contradiction. If not, there are $\varepsilon_0 > 0$ and $r_l \downarrow 0$ such that

$$\Sigma(u_{r_l}) \cap \{y \in B_1 : \text{dist}(y, \Sigma(q)) > \varepsilon_0\} \neq \emptyset \quad \text{for all } q \in S_k^{\text{even}}.$$

Thanks to Proposition 5.12, up to a subsequence, we have

$$\frac{(u - P_k)_{r_l}}{H(r_l, u - P_k)^{1/2}} \rightarrow \bar{q} \in S_k^{\text{even}} \setminus \{0\}, \quad \text{in } C_{\text{loc}}^0(\mathbb{R}^n)$$

Thus, there are $y_l \in B_1$ such that

$$y_l \in \Sigma(u_{r_l}), \quad \text{dist}(y_l, \Sigma(\bar{q})) \geq \varepsilon_0, \quad y_l \rightarrow y_\infty \in \{x_n = 0\}$$

By Lemma 3.1, we can write $\bar{q}(x) = -|x_n|(q_0(x') + x_n^2 q_1(x))$ for some polynomials q_0, q_1 , with $q_0 \geq 0$. For brevity, we set $h_l := H(r_l, u - P_k)^{\frac{1}{2}}$, then $r_l^{k+\delta} \lesssim h_l \ll r_l^k$. As $y_l \rightarrow y_\infty$, in order to reach for a contradiction it suffices to show that for some small radius $R > 0$ and all l large

$$\Sigma(u_{r_l}) \cap B_R(y_\infty) = \emptyset \tag{9}$$

The rest of the proof is devoted to showing (9) for a suitable R independent on l .

We start choosing a radius ρ as follows. As $y_\infty \in \{x_n = 0\} \setminus \Sigma(\bar{q}) \subseteq \{q_0 > 0\}$ we can find some small $\rho, m \in (0, 1)$ such that

$$q(x) \leq -m|x_n| \quad \text{for } x \in B_{4\rho}(y_\infty).$$

And we can also require that $\rho \leq m/100$.

Let us introduce some notations. Define the set of points that admit a barrier

$$Z_l := \left\{ z \in B_\rho(y_\infty) : \exists \phi_{z,l} \text{ of class } C^2 \text{ in a neighborhood of } \overline{B_\rho(z)} \text{ solving (10)} \right\},$$

where

$$\begin{cases} \phi_{z,l}(z) = 0, \\ \phi_{z,l} \geq 0, \\ \Delta \phi_{z,l} < r_l^2 & \text{in } \overline{B_\rho(z)}, \\ u_{r_l} < \phi_{z,l} & \text{on } \partial B_\rho(z). \end{cases} \quad (10)$$

We also set

$$\Gamma_l := \{\mathcal{A}_k(r_l \cdot) = 0\} \cap B_2$$

For l large in terms of n, k and τ , Γ_l is a smooth hypersurface inside B_2 , which converges to the hyperplane $\{x_n = 0\}$. Furthermore, combining Lemma 3.10 and the fact that $\mathcal{A}_k(x) = x_n + O(|x|^2)$, we get that

$$(\{u_{r_l} = 0\} \cup \Gamma_l) \cap B_2 \subseteq \{|x_n| \leq C_1 r_l^{\alpha_\circ}\} \cap B_2$$

for some constant $C_1 = C_1(n, k) > 0$.

Claim. There is l_0 such that for $l > l_0$ the following holds.

- (a) All points in $B_\rho(y_\infty)$ belonging to Γ_l admit a barrier, that is $\Gamma_l \cap B_\rho(y_\infty) \subseteq Z_l$.
- (b) The function u vanishes on Z_l . Moreover, Z_l is open and contained in the interior of the contact set $\{u(r_l \cdot) = 0\}$.
- (c) There is a dimensional constant $N(n) > 0$ such that $(\Sigma(u_{r_l}) \cap B_{\rho/(2N)}(y_\infty)) \setminus \Gamma_l = \emptyset$.

The combination of these three claims will give (9) with $R = \rho/(2N)$. In fact, as there are no singular points in the interior of the singular set, (a) and (b) give $\Sigma(u_{r_l}) \cap B_{\rho/(2N)}(y_\infty) \cap \Gamma_l = \emptyset$. Then we can conclude with (c). We end with the proof of this claim.

Proof of the Claim. The proof of (a) is lengthy, and independent of (b) and (c). So we assume (a) holds at first. We begin by proving (b). First, for any $z \in Z_l$ and any ξ close to z we can define

$$\phi_{\xi,l}(x) := \phi_{z,l}(x + (z - \xi)).$$

By continuity of translations $\phi_{\xi,l}$ solves (10) in $B_\rho(\xi)$, for $|\xi - z|$ small enough, hence Z_l is open. We now apply the comparison principle, using to the last two properties of the barrier in (10) to find $u_{r_l}|_{Z_l} \equiv 0$. Notice that for $z \in Z_l$, two cases arise: either for all $c > 0$ we have $u(r_l \cdot) < \phi_{z,l} + c$ on $B_\rho(z)$, or there exist the largest $c_* > 0$ such that $u(r_l x_*) = \phi_{z,l}(x_*) + c_*$ for some $x_* \in \overline{B_\rho(z)}$. In the first case evaluating at z and sending $c \downarrow 0$ we get $u(r_l z) = 0$. In the second case we notice that by (10) we must have $x_* \notin \partial B_\rho(z)$, thus we get

$$r_l^2 \chi_{\{u(r_l \cdot) > 0\}}(x_*) = \Delta u_{r_l}(x_*) \leq \Delta \phi_{z,l}(x_*) < r_l^2 \quad \Rightarrow \quad x_* \notin \{u(r_l \cdot) > 0\}.$$

Then $0 = u(r_l x_*) = \phi_{z,l}(x_*) + c_* \geq c_* > 0$, a contradiction.

Next, we turn to the proof of (c). First recall that there exists a modulus of continuity such that for all $x \in \Sigma(u)$ we have

$$\|u(x + \cdot) - p_{2,x}\|_{L^\infty(B_r)} \leq r^2 \omega(r).$$

Suppose by contradiction that we can find

$$y_l^* \in (\Sigma(u_{r_l}) \cap B_{\rho/2N}(y_\infty)) \setminus \Gamma_l$$

for arbitrarily large l . We set for brevity $p_l := p_{2,r_l y_l^*}$. As p_l is convex and $\Delta p_l \geq 1$, for every such l we choose a unit vector e_l such that

$$p_l(x) \geq \frac{1}{2n} (e_l \cdot x)^2$$

Now by (a) and (b), we have for all $z \in B_{\rho/N}(y_\infty) \cap \Gamma_l$

$$\begin{aligned} 0 &= u(r_l z) = u_{r_l}(y_l^* + (z - y_l^*)) \\ &\geq p_l(z - y_l^*) - |z - y_l^*|^2 \omega(|z - y_l^*|) \\ &\geq \frac{1}{2n} (e_l \cdot (z - y_l^*))^2 - |z - y_l^*|^2 \omega(|z - y_l^*|). \end{aligned}$$

We reach a contradiction from this: on the one hand we have $|z - y_l^*| \leq 1/N$, so we can require N so large that $\omega(|z - y_l^*|) \leq 1/100n$. On the other hand we can choose z in such a way that the nonzero vector $z - y_l^*$ is almost aligned with e_l (because $p_l \rightarrow p_0$, we can let $e_l \rightarrow \{p_2 = 0\}$), thus we get a contradiction dividing by $|z - y_l^*|^2$.

Now we turn to (a). We need to construct a barrier for all $z \in B_\rho(y_\infty) \cap \Gamma_l$. Set

$$\phi_{z,l}(x) := \left(1 - \frac{h_l}{r_l^2}\right) \frac{1}{2} \mathcal{A}_k^2(r_l x) + \frac{h_l}{4n} |x' - z'|^2$$

We have to check that $\phi_{z,l}$ indeed satisfies (10) for l large. The first two equations in (10) are clearly fulfilled for any l . For the third condition, let us compute for $x \in \overline{B_\rho(z)}$:

$$\Delta \phi_{z,l} = r_l^2 - h_l + C_2 r_l^{k+2} + \frac{2(n-1)}{4n} h_l \leq r_l^2 - \frac{1}{2} h_l + C_0 C_2 r_l h_l,$$

where we used that, for some $C_2 = C_2(n, k)$, we have $\Delta(\frac{1}{2} \mathcal{A}_k^2) \leq 1 + C_2 |x|^k$ and $r_l^{k+1} \leq C_0 h_l$. Hence, $\Delta \phi_{z,l} < r_l^2$, as soon as l is large enough. We turn to the last condition of (6.5). For any fixed $\eta \leq \frac{\rho^2}{100n}$, we have by uniform convergence, for l large

$$u_{r_l} \leq \frac{1}{2} \mathcal{A}_k^2(r_l \cdot) + h_l q + h_l \eta$$

in B_2 . As for some constant $C_3 = C_3(n, k) > 0$ we have in B_2

$$\frac{1}{2} \mathcal{A}_k^2(x) \leq \frac{1}{2} x_n^2 + C_3 |x|^3$$

recalling the choice of ρ , we get for $x \in \overline{B_\rho(z)}$:

$$\begin{aligned} u(r_l x) &\leq \frac{1}{2} \mathcal{A}_k^2(r_l x) - h_l m |x_n| + h_l \eta \\ &\leq \left(1 - \frac{h_l}{r_l^2}\right) \frac{1}{2} \mathcal{A}_k^2(r_l x) + \frac{h_l}{2} x_n^2 - h_l m |x_n| + C_3 h_l r_l |x|^3 + h_l \eta \\ &= \phi_{z,l}(x) + h_l \left(\frac{1}{2} x_n^2 - m |x_n| + C_3 r_l |x|^3 + \eta - \frac{1}{4n} |x' - z'|^2 \right). \end{aligned}$$

We show that whenever $x \in \partial B_\rho(z)$ the term between parentheses is negative. In fact, recalling $|z_n| \leq C_1 r^{\alpha_0}$ and $x \in B_2$,

$$\begin{aligned} &\frac{1}{2} x_n^2 - m |x_n| + C_3 r_l |x|^3 + \eta - \frac{1}{4n} |x' - z'|^2 \\ &= \frac{1}{4n} |x_n - z_n|^2 + \frac{1}{2} x_n^2 - m |x_n| + C_3 r_l |x|^3 + \eta - \frac{\rho^2}{4n} \\ &\leq x_n^2 - m |x_n| + C_1^2 r_l^{2\alpha_0} + 8C_3 r_l + \eta - \frac{\rho^2}{4n}. \end{aligned}$$

We claim that this quantity is negative as soon as $r_l^{\alpha_0} \leq \min \left\{ \frac{\rho}{C_1}, \frac{\eta}{8C_3}, \frac{\sqrt{\eta}}{C_1}, 1 \right\}$. In fact,

$$|x_n| \leq |x_n - z_n| + |z_n| \leq \rho + C_1 r_l^{\alpha_0} \leq 2\rho \leq \frac{2m}{100}$$

thus $x_n^2 - m |x_n| \leq 0$ and

$$C_1^2 r_l^{2\alpha_0} + 8C_3 r_l + \eta - \frac{\rho^2}{4n} \leq 3\eta - \frac{\rho^2}{4n} \leq \frac{3\rho^2}{100n} - \frac{\rho^2}{4n} < 0.$$

This completes the proof. $\square$

Corollary 6.7. *Let $k \geq 3$ be odd, then $\dim_{\mathcal{H}}(\Sigma^{kth} \setminus \Sigma^{>k}) \leq n - 2$. Moreover, if $n = 2$, then $\Sigma^{kth} \setminus \Sigma^{>k}$ is discrete in Σ (hence countable).*

Proof. $\dim_{\mathcal{H}}(\Sigma^{kth} \setminus \Sigma^{>k}) \leq n - 2$ follows by Lemma 6.2 (with τ a constant) and Lemma 6.6. Moreover, if $n = 2$, then $\Sigma(q) = \{0\}$ for any $0 \neq q \in S_k^{\text{even}}(L)$ (as such q has an explicitly formula and one can check it by direct computation). Take $\varepsilon = 1/2$ in Lemma 6.6, then for any $x \in \Sigma^{kth} \setminus \Sigma^{>k}$, we have $\Sigma \cap (B_r(x) \setminus B_{r/2}(x)) = \emptyset$ if $r \ll 1$. $\square$

6.3 The size of $\Sigma^{>k} \setminus \Sigma^{\geq k+1}$

We verify a lemma which is similar to Lemma 3.11.

Lemma 6.8. *Let $0 \in \Sigma^{>k} \setminus \Sigma^{\geq k+1}$. Suppose $r_l \rightarrow 0$, $0 \neq x_l \in \Sigma^{kth} \cap B_{r_l}$ and $\lambda_k(x_l) \rightarrow \lambda_k$. Assume further that as $l \rightarrow \infty$, $x_l/r_l \rightarrow y_\infty \in \overline{B_1}$, and $\tilde{v}_{r_l} := (u - P_k)_{r_l}/H(r_l, u - P_k)^{1/2} \rightarrow q$ in $C_{loc}^0(\mathbb{R}^n)$ for some $0 \neq q \in S_{\lambda_k}(\{p_2 = 0\})$. Then $y_\infty \in \{p_2 = 0\}$ and $q(y_\infty + \cdot) = q$.*

Proof. $y_\infty \in \{p_2 = 0\}$ holds because $r_l^{-2}u(r_l \cdot) \rightarrow p_2$ locally uniformly. Up to a subsequence,

$$\tilde{v}_{x_l, r_l} := \frac{u(x_l + r_l \cdot) - P_{k, x_l}(r_l \cdot)}{\left\| (u(x_l + \cdot) - P_{k, x_l})_{r_l} \right\|_{L^2(\partial B_1)}} \rightarrow Q$$

in $C_{\text{loc}}^0(\mathbb{R}^n)$, for some $Q \in \mathcal{S}_{\lambda_k}(\{p_2 = 0\}) \setminus \{0\}$. On the other hand, by uniform convergence,

$$q(y_\infty + \cdot) = \lim_l \tilde{v}_{r_l}(x_l/r_l + \cdot) = \lim_l \frac{u(x_l + r_l \cdot) - P_k(x_l + r_l \cdot)}{H(r_l, u - P_k)^{1/2}}.$$

We can write

$$\tilde{v}_{r_l}(x_l/r_l + \cdot) = \tilde{v}_{r_l, x_l} \cdot I_l + J_l$$

where

$$I_l := \frac{H(r_l, u(x_l + \cdot) - P_{k, x_l}(\cdot))^{1/2}}{H(r_l, u - P_k)^{1/2}} = \frac{\|u(x_l + r_l \cdot) - P_{k, x_l}(r_l \cdot)\|_{L^2(\partial B_1)}}{\|u(r_l \cdot) - P_k(r_l \cdot)\|_{L^2(\partial B_1)}}$$

is a numerical sequence and

$$J_l := \frac{P_{k, x_l}(r_l \cdot) - P_k(x_l + r_l \cdot)}{H(r_l, u - P_k)^{1/2}}$$

is a sequence of harmonic polynomials of degree at most $k + 1$. Now two cases arise:

$$\text{either } \sup_l I_l < \infty, \text{ or } I_{l_m} \uparrow \infty \text{ for some subsequence } l_m \rightarrow \infty.$$

Let us begin with the first case. Up to a subsequence that we do not rename, we have $I_l \rightarrow \alpha$ for some $\alpha \geq 0$. And $J_l \rightarrow J$ locally uniformly to some harmonic polynomial J of degree at most $k + 1$. Thus, sending $l \uparrow \infty$ in (6.7), we obtain

$$q(y_\infty + \cdot) = \alpha Q + J.$$

We exploit homogeneity: for large $R > 0$ it holds

$$R^{\lambda_k} q\left(\frac{y_\infty}{R} + \cdot\right) = R^{\lambda_k} \alpha Q + J(R \cdot),$$

so $\lim_{R \uparrow \infty} R^{-\lambda_k} J(Rx)$ exists for every $x \in \mathbb{R}^n$. As λ_k is not an integer and J is a polynomial, the only possibility is that $\lim_{R \uparrow \infty} R^{-\lambda_k} J(Rx) = 0$ for all x and so $\deg J \leq k$. Hence the last identity reads

$$q = \alpha Q.$$

Inserting this back, we find $q(y_\infty + \cdot) = q + J$. Now, using again that q is homogeneous, for any $R > 0$ we have

$$R \left(q\left(\frac{y_\infty}{R} + \cdot\right) - q(\cdot) \right) = R^{1-\lambda_k} J(R \cdot)$$

Sending $R \rightarrow \infty$ the left hand side converges to $y_\infty \cdot \nabla q$, but as before the right hand side can only converge to 0, so $y_\infty \cdot \nabla q = 0$, which means $q(y_\infty + \cdot) = q$.

The second case is simpler. We divide equation (6.7) by I_{l_m} and find, after passing to a subsequence of l_m , that

$$0 = Q + \tilde{J}$$

for some harmonic polynomial $\tilde{J}$ of degree at most $k + 1$. This is a contradiction, since $Q \neq 0$ is a $\lambda_k \in (k, k + 1)$ homogeneous function, hence not a polynomial. This finishes the proof. $\square$

Lemma 6.9. *For all $x \in \Sigma^{kth} \setminus \Sigma^{(k+1)th}$ and for all $\varepsilon > 0$, there exist $\rho = \rho(x, \varepsilon) > 0$ and a $(n-2)$ -dimensional plane $\Pi_{x,r}$ passing through x , such that $\forall r \in (0, \rho)$,*

$$E \cap B_r(x) \cap \lambda_k^{-1}([\lambda_k(x) - \rho, \lambda_k(x) + \rho]) \subseteq \{x : \text{dist}(x, \Pi_{x,r}) \leq \varepsilon r\},$$

where λ_k denotes the function $\Sigma^{kth} \setminus \Sigma^{(k+1)th} \ni x \mapsto \lambda_k(x)$.

Proof. We argue by contradiction. Assume for $x = 0$ and some $\varepsilon > 0$ the above doesn't hold, then for each l , there exists $r_l \in (0, 2^{-l})$ and $n-1$ points $x_l^{(1)}, \dots, x_l^{(n-1)} \in E \cap B_{r_l}$, such that for $j = 1, \dots, n-1$,

$$|\lambda_k(x_l^{(j)}) - \lambda_k| \leq 2^{-l} \quad \text{and} \quad \text{dist}(x_l^{(j)}, \text{span}\{x_l^{(1)}, \dots, x_l^{(j-1)}\}) \geq \varepsilon r_l.$$

For each fixed j , the sequence $x_l^{(j)}$ lies in $E \subseteq \Sigma^{kth}$, up to a subsequence (and by diagonal method) we can assume $x_l^{(j)}/r_l \rightarrow x_\infty^{(j)}$. Then $y_\infty^{(1)}, \dots, y_\infty^{(n-1)}$ still spans a $(n-1)$ -dimensional plane. We have

$$\tilde{v}_l := \frac{(u - P_k)_{r_l}}{H(r_l, u - P_k)^{1/2}} \rightarrow q,$$

for some $q \in S_{\lambda_k}(p_2 = 0)$. By Lemma 6.8, q is translation-invariant in the directions $y_\infty^{(j)}$ for all $j = 1, \dots, n-1$. Hence q is 1-dimensional solution of the obstacle problem, which can easily seen to have the form $q(x) = -A|x_n| + Bx_n$, contradictory to $\lambda_k \geq k > 1$. $\square$

Corollary 6.10. *Let $k \geq 2$, then $\dim_{\mathcal{H}}(\Sigma^{>k} \setminus \Sigma^{\geq k}) \leq n-2$. Moreover, if $n = 2$, then $\Sigma^{>k} \setminus \Sigma^{\geq k+1}$ is empty if k is even, and is discrete (hence countable) in $\Sigma(u)$ if k is odd.*

Proof. The dimension estimate follows by Lemma 6.3 and Lemma 6.9. It remains the case $n = 2$. Recall that if q is a λ -homogeneous solution to the thin obstacle problem, then $\lambda \in \mathbb{N}_+ \cup \{2m - \frac{1}{2} : m \in \mathbb{N}_+\}$ (see Lemma 3.1). Thus we find that $\Sigma^{>k} \setminus \Sigma^{\geq k+1}$ is empty for k even. If k is odd, we find $\lambda_k(x_o) = k + \frac{1}{2}$ for every $x_o \in \Sigma^{>k} \setminus \Sigma^{\geq k+1}$. If this set is not discrete, we could apply Lemma 6.8 and obtain a one-dimensional and $(k + \frac{1}{2})$ -homogeneous solution of the thin obstacle problem, which is impossible. $\square$

6.4 The size of $\Sigma^{\geq k+1} \setminus \Sigma^{(k+1)th}$

Lemma 6.11. *Let $k \geq 2$, $x \in \Sigma^{\geq k+1} \setminus \Sigma^{(k+1)th}$ and $\varepsilon > 0$, then there exists $\rho(\varepsilon, x) > 0$, such that for each $r \in (0, \rho)$, there exists $0 \neq q \in S_{k+1}^{\text{even}}(\{p_{2,x} = 0\})$ such that*

$$\Sigma(u) \cap B_r(x) \subseteq \Sigma(q) + B_{\varepsilon r}(x).$$

Proof of the case $(k+1)$ even. Up to an isometry we can assume $x = 0$ and $p_2 = \frac{1}{2}x_n^2$. We argue by contradiction. Otherwise there exists $\varepsilon > 0$ and a sequence $r_l \rightarrow 0$ such that

$$y_l \in \Sigma(u_{r_l}) \cap B_1 \quad \text{and} \quad \text{dist}(y_l, \Sigma(q')) \geq \varepsilon$$

for all $q' \in S_{k+1}^{\text{even}}(\{x_n = 0\})$. We can assume $y_l \rightarrow y_\infty \in \{x_n = 0\}$ and (since $H_{k+1}(0+, u - P_k)$ exists by Lemma 5.3)

$$\frac{(u - P_k)(r_l \cdot)}{r_l^{k+1}} \rightarrow \bar{q} \in \mathcal{S}_{k+1}(\{x_n = 0\})$$

in $C_{\text{loc}}^0(\mathbb{R}^n)$. Since $0 \notin \Sigma^{(k+1)\text{th}}$ we have $\bar{q}^{\text{even}} \neq 0$. We can rewrite this as

$$w_l := \frac{\left(u - \frac{1}{2}\mathcal{A}_{k+1}^2(p_{2,0}, \dots, p_{k,0}, \bar{q}^{\text{odd}})\right)_{r_l}}{r_l^{k+1}} \rightarrow \bar{q}^{\text{even}}, \quad \text{in } C_{\text{loc}}^0(\mathbb{R}^n).$$

Now recall that, $k+1$ being even, we have $\Sigma(\bar{q}^{\text{even}}) = \{\bar{q}^{\text{even}} = 0\}$, thus $\eta := \bar{q}^{\text{even}}(y_\infty) > 0$ as y_∞ lies on the thin obstacle. So we can find a small radius $\delta > 0$ and a large l_0 such that for all $l > l_0$ we have

$$\inf_{B_\delta(y_\infty)} w_l \geq \inf_{B_\delta(y_\infty)} \bar{q}^{\text{even}} - \|w_l - \bar{q}^{\text{even}}\|_{L^\infty(B_2)} \geq \frac{\eta}{2} - \frac{\eta}{4} = \frac{\eta}{4}$$

However, eventually we will have $y_l \in B_\delta(y_\infty)$ and this is a contradiction as

$$0 \geq -\frac{\mathcal{A}_{k+1}^2(r_l y_l)}{2r_l^{k+1}} = w_l(y_l) \geq \inf_{B_\delta(y_\infty)} w_l \geq \eta/4 \quad \square$$

We remark that in this proof we only used that $y_l \in \{u_{r_l} = 0\}$, not that y_l were singular points. We've actually proved that, if $(k+1)$ is even, then the statement can be strengthened as

$$\{u = 0\} \cap B_r(x) \subseteq \Sigma(q) + B_{\varepsilon r}(x).$$

Proof of the case $(k+1)$ odd. Arguing by contradiction as in the even case we find $\varepsilon > 0$ and a sequence $r_l \rightarrow 0$ such that for each l

$$y_l \in \Sigma \cap B_1 \text{ and } \text{dist}(y_l, \Sigma(q)) \geq \varepsilon_0, \quad \forall q \in \mathcal{S}_{k+1}^{\text{even}}(\{x_n = 0\}) \setminus \{0\}.$$

We can also assume that $y_l \rightarrow y_\infty \in \{x_n = 0\}$ and

$$w_l := \frac{\left(u - \frac{1}{2}\mathcal{A}_{k+1}^2(p_{2,0}, \dots, p_{k,0}, \bar{q}^{\text{odd}})\right)_{r_l \cdot}}{r_l^{k+1}} \rightarrow \bar{q}^{\text{even}}, \quad \text{in } C_{\text{loc}}^0(\mathbb{R}^n).$$

From this point the proof is conducted analogously to the proof of Lemma 6.6, it suffices to replace k with $k+1$ and h_l with r_l^{k+1} . $\square$

We conclude exactly the same as Corollary 6.7.

Corollary 6.12. *Let $k \geq 2$, then $\dim_{\mathcal{H}}(\Sigma^{\geq k+1} \setminus \Sigma^{(k+1)\text{th}}) \leq n-2$. Moreover, if $n = 2$, then $\Sigma^{\geq k+1} \setminus \Sigma^{(k+1)\text{th}}$ is discrete in Σ (hence countable).*

6.5 Proof of Theorem 1.1(1) and Theorem 1.4(1)

This subsection is devoted to prove Theorem 1.1(1) and Theorem 1.4. The first one is fairly easy.

Lemma 6.13. $\dim_{\mathcal{H}}(\Sigma^{2nd} \setminus \Sigma^{>3}) \leq n-3$. And when $n = 3$, this set is discrete in Σ_{n-1} .

Proof. We start with the case $n = 3$. Suppose on the contrary that $0 \in \Sigma_2^a = \Sigma^{>2} \setminus \Sigma^{\geq 3}$ (thanks to Proposition 3.2, and $\Sigma_{n-1} \ni x_l \rightarrow 0(x_l \neq 0)$). Let $r_l = |x_l|$ and suppose $x_l/r_l \rightarrow y_\infty$, then the blow-up limit $(u - p_2)_{r_l}/H(r_l, u - p_2)^{1/2} \rightarrow q$ satisfies $q(y_\infty + \cdot) = q(y_\infty)$ by Lemma 6.8. Now q only depends on two variables, and λ_2 -homogeneous, by Lemma 3.1 we have $\lambda_2 \geq 3$. Contradictory!

For $n \geq 4$, we can consider the accumulation set $\mathcal{A}$ (c.f. Lemma 3.15). It's easy to see $\mathcal{A} \subseteq \{p_2 = 0\} \cap \{q = 0\}$, where $q \in S_{\lambda_2}(\{p_2 = 0\})$ and $\lambda_2 \in [2 + \alpha_0, 3)$. Furthermore, according to the definition of accumulation set, we obtain as above that for any $z \in \mathcal{A}$, we have $\gamma(0+, q(z + \cdot)) \in [2 + \alpha_0, 3)$. Lemma 3.1 implies such set has Hausdorff dimension $\leq n-3$. And we are done. $\square$

Proof of Theorem 1.1(1). Since $\lambda_2(x) \geq 2 + \alpha_0$ for $x \in \Sigma_{n-1}$, we obtain Σ_{n-1} can be covered by C^{1,α_0} hypersurface by Lemma 3.19. In the same manner, Σ_{n-1}^g is relatively closed in Σ_{n-1} and covered by $C^{1,1}$ hypersurface. The dimension statement is Lemma 6.13. Finally, when $n = 2$, we've showed in Lemma 3.9 that $\Sigma_1 = \Sigma_1^g$. $\square$

Now we turn to Theorem 1.4, which is a easy corollary of what we've known and the C^∞ Whitney's extension Theorem:

Lemma 6.14. *Let $E \subseteq \mathbb{R}^n$ be any set. Suppose for each $x_0 \in E$ and $k \in \mathbb{N}$, there exists a polynomial Q_{k,x_0} of degree at most k , such that*

- (1) $Q_{k,x_0} = \pi_{\leq k}(Q_{k+l,x_0})$, for all $k, l \in \mathbb{N}$ and $x_0 \in E$;
- (2) for each $k \in \mathbb{N}$, there's a constant $C(k)$ such that for all multi-index α with $|\alpha| \leq k$, we have

$$|\partial^\alpha Q_{k,x_0}(0) - \partial Q_{k,x}(x_0 - x)| \leq C(k)|x - y|^{k-|\alpha|+1}.$$

Then there exists a C^∞ function $F: \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$F(x_0 + h) = Q_{k,x_0}(h) + O(|h|^{k+1}), \quad \text{as } |h| \rightarrow 0.$$

Set $\Sigma^\infty := \cup_{k \geq 2} \Sigma^{kth}$. We extract the main issue of Theorem 1.4 as a lemma.

Lemma 6.15. Σ^∞ is closed and locally covered by a smooth manifold of dimension $n - 1$.

Proof. Assume $0 \in \Sigma^\infty$. We want to apply Lemma 6.14 with $E = \Sigma^\infty \cap B_{1/8}$ and the polynomials

$$Q_{k,x} := \pi_{\leq k}(P_{k,x}) \quad \text{for all } x \in \Sigma^\infty \cap B_1, k \geq 0.$$

Assumption (1) is holds because $P_{k+l,x}$ and $P_{k,x}$ agree up to order k . We need to show that (2) holds, it is not restrictive to do it only for some fixed $k \geq 3$. One can mimic the proof of Theorem 1.1(2) as before, to obtain:

$$\|u(x + \cdot) - P_{k,x}\|_{L^\infty(B_r(0))} \leq Cr^{k+1}.$$

The only difference is we can now get the L^∞ (rather than L^2) control due to Lemma 4.7. Let $x_1, x_2 \in \Sigma^\infty \cap B_{1/8}$ and $r = |x_1 - x_2|$, then

$$\begin{aligned} & \|Q_{k,x_1}(\cdot - x_1) - Q_{k,x_2}(\cdot - x_2)\|_{L^\infty(B_r(x_1))} \\ & \leq \|u(\cdot) - P_{k,x_1}(\cdot - x_1)\|_{L^\infty(B_r(x_1))} + \|u - P_{k,x_2}(\cdot - x_2)\|_{L^\infty(B_r(x_1))} + Cr^{k+1} \\ & \leq \|u(x_1 + \cdot) - P_{k,x_1}(\cdot)\|_{L^\infty(B_r(x_1))} + \|u(x_2 + \cdot) - P_{k,x_2}\|_{L^\infty(B_{2r}(x_2))} + Cr^{k+1} \\ & \leq Cr^{k+1}. \end{aligned}$$

We have checked (2) since norms are equivalent. By Whitney extension theorem there exists a C^∞ function $F: \mathbb{R}^n \rightarrow \mathbb{R}$ for which

$$F(x) = P_{k,x_0}(x - x_0) + O(|x - x_0|^{k+1})$$

at every $x_0 \in \Sigma^\infty \cap \overline{B_1}$. We now conclude as before, as $\Sigma^\infty \subseteq \{\nabla F = 0\}$ and $\nabla^2 F(x_1) = \nabla^2 p_{2,x_1}(0)$ has rank 1.

Finally, let us now prove that Σ^∞ is closed. Suppose $0 \in \overline{\Sigma^\infty}$ and so there is $x_l \in \Sigma^\infty$ such that $x_l \rightarrow 0$. First observe that $0 \in \Sigma$ as the full singular set Σ is closed, hence p_2 exists. We define by continuity $P_k := \lim_l P_{k,x_l}$ for all $k \geq 3$. This is well-defined as we've showed above that the map $x \mapsto P_{k,x}$ is Lipschitz on Σ^∞ , hence admits a canonical extension to the closure. For some constants $C, \varepsilon > 0$ and a radius r_0 , both independent from l , it holds for all $r \in (0, r_0)$,

$$\frac{d}{dr} (r^{-2k} H(r, u(x_l + \cdot) - P_{k,x_l})) \geq -Cr^{\varepsilon-1} \quad \text{and} \quad r^{-2k} H(r, u(x_l + \cdot) - P_{k,x_l}) \leq C.$$

We can pass both these inequalities to the limit $l \rightarrow \infty$ and show that the sequence $r^{-k}(u - P)_r$ is uniformly bounded in $H_{\text{loc}}^1(\mathbb{R}^n)$. Hence it is immediate that $0 \in \tilde{\Sigma}^{(k-1)\text{th}}$ (see Lemma 5.8), as k was arbitrary we conclude $0 \in \Sigma^\infty$. $\square$

Proof of Theorem 1.4. The bound on the dimension of $\Sigma \setminus \Sigma^\infty$ is given in previous subsections. We are done by Lemma 6.15. $\square$

7 Generic regularity in $\mathbb{R}^3$ and $\mathbb{R}^4$

References

- [1] L. Caffarelli, *The obstacle problem revisited*, J. Fourier Anal. Appl., **4** (1998), 383-402.
- [2] A. Figalli, X. Ros-Oton, J. Serra, *Generic regularity of free boundaries for the obstacle problem*, Publ. Math. IHÉS **132** (2020), 181-292.
- [3] A. Figalli, J. Serra, *On the fine structure of the free boundary for the classical obstacle problem*, Invent. Math. **215** (2019), 311-366.
- [4] F. Franceschini, W. Zatoń, *C^∞ partial regularity of the singular set in the obstacle problem*, preprint (2021).
- [5] A. Naber, D. Valtorta, *Volume estimates on the critical sets of solutions to elliptic PDEs*, Comm. Pure Appl. Math. **70** (2017), 1835-1897.
- [6] D. Schaeffer, *Some examples of singularities in a free boundary*, Ann. Scuola Norm. Sup. Pisa **4** (1976), 131-144.
- [7] L. Simon, *Introduction to geometric measure theory*, available at <https://web.stanford.edu/class/math285/ts-gmt.pdf>.
- [8] R. Monneau, *On the number of singularities for the obstacle problem in two dimensions*, J. Geom. Anal. **13** (2003), 359-389.